

- ii. Fuel stores should comply with local dangerous goods laws and all combustible materials are to be kept clear of buildings Industrial LPG stores should be external to occupied structures
- iii. Hazardous materials e.g diesel should be stored in a bunded area and spill kits provided  
Clear and multilingual signage should be provided around hazardous storage areas

### **15.23 Fire Safety and Fire Fighting**

- i. Open fires/burning should be strictly prohibited
- ii. Fire prevention and firefighting /control facilities should be provided, particularly in food preparation areas
- iii. Fire fighting hoses should be engineered to provide full coverage of camp perimeter and be capable of delivering the appropriate water pressure to fight fire
- iv. All structures should be constructed with fire resistant material (metal or fire retardant treated wood form)
- v. Fire and smoke alarms should be installed in all accommodation areas
- vi. Where possible, buildings should be separated from each other by providing space or fire resistant material

### **15.24 Mechanical and Electrical Safety**

- i. Electrical equipment and cords should be maintained and physically protected to prevent the risk of electrocution or shock
- ii. Mechanical equipment should be operated by trained personnel only
- iii. Electrical circuit breakers should be fitted to all distribution boards
- iv. Electrical register of equipment should be maintained and testing to be undertaken and recorded by qualified Electrical Engineer

### **15.25 Housekeeping**

- i. Waste should be centrally located in a segregated, easily cleaned area, kept in rodent and fly tight containers, collected regularly and grounds are to be kept free from rubbish, debris and other refuse
- ii. Effective housekeeping measures should be taken to ensure pests are not attracted to the site eg removal of all food sources and breeding areas.
- iii. No pets, birds or livestock should be kept.
- iv. Areas should be kept free of hazards e.g slips, trip or fall hazards

### **15.26 Transportation**

- i. The transport facilities are to be provided to site, these should be safe, with trained drivers and in accordance with local laws

### **15.27 Pest Control**

- i. In mosquito prone areas, checks for mosquito breeding, including the presence of larvae in all stagnant water within the worker's quarter's premises should be carried out at least once a week. Thermal fogging of worker's quarters premises should be carried out at least once each fortnight.
- ii. Adequate controls should also to be in place to prevent pest infestations e.g inspections and preventative measures such as baits and other controls

### **15.28 Camp / Accommodation Supervisor**

- i. A camp supervisor should be appointed to coordinate all activities, communication and inspections etc of the camp. This supervisor should speak at least two of the languages spoken by the workforce.
- ii. A team comprising of contractor's Labour Welfare Officer and Chief OHS&E Manager shall conduct a monthly labour colony Safety Audit and the report shall be submitted no later than the 7th of each month within the Contractor's monthly OH&S Report.

## **16. Working at Height**

### **16.1 General**

- 16.1.1 Working at height is the largest single cause of serious accidents in the construction industry and therefore the Contractor shall carry out risk assessments for all work where workers or materials can fall more than two metres. The contractors will ensure that no height work will be undertaken unless a Permit to Work, has been prepared and issued.
- 16.1.2 Where work is being carried out above areas where there is public access such as roads footpaths etc. particular care must be taken to ensure that no materials can fall from the working area.
- 16.1.3 Edge protection shall be provided at all leading edges or openings where workers or materials can fall more than two meters. Edge protection shall meet the minimum standard of;
  - a. A main guardrail at least 1 meter above the edge
  - b. A toe board at least 200 mm high; and
  - c. An intermediate guard rail or other barrier so that there is no gap more than 470 mm.
  - d. Rebar shall not be used as edge protection only standard pipes should be used.
  - e. The pipes shall be painted in reflective paints.

### **16.2 Use of Scaffolds**

- 16.2.1 All scaffolds should be erected and dismantled by workmen and Inspected by the scaffold supervisor who are thoroughly trained by third Party agency and the training requirements should meet the construction industry training board(CITB) UK, Construction Industry scaffolders records scheme (CISRS) UK or Scaffolding Training

Institute, USA. The Profile of the training Agency, ISO certifications, the name, educational qualifications and experience for of the trainers are to be submitted to the Employer/Engineer a minimum 28 days before prior to appointment of the agency. Only upon notice of no objection by the Employer /Engineer shall be authorised to work on CMRL site.

- 16.2.2 All scaffolds should be inspected by a competent person at least every day after erection and the results of inspections recorded and the records shall be kept available for checking by the Engineer.
- 16.2.3 The Contractor shall develop a scaffold tagging system acceptable to the Employer /Engineer to indicate:
- a) Scaffolds under construction or demolition;
  - b) Scaffolds that are complete but have hazards associated with them; and
  - c) Scaffolds erected and safe for use.
- 16.2.4 Tags shall be fitted to all scaffolds to show whether they are safe for use or not. All Safe for Use tags shall be signed by a third party trained scaffold inspector from the contractor.
- 16.2.5 The Contractor shall envelope all scaffolds with screen nets to prevent debris from falling outside the scaffold.
- 16.2.6 Contractor shall use only proprietary access ladders and working platforms for system formworks. No mix and match using conventional catwalks and monkey ladders is allowed.
- 16.2.7 All scaffolds shall be constructed of sound materials free from patent defect.
- 16.2.8 The following measures shall be taken;
- (a) the scaffold shall be constructed for the correct use (Light or Heavy Duty)
  - (b) securely fixed to existing structures or adequately buttressed;
  - (c) the use of barrels, boxes, loose tiles or other unsuitable material shall not be used as supports for working platforms;
  - (d) all working platforms shall be fully boarded;
  - (e) all working platforms shall have guard rails at one metre height and shall also have an intermediate rail at half height;
  - (f) all working platforms shall be provided with toe boards;
  - (g) all working platforms shall be kept free of unnecessary obstruction or rubbish
  - (h) secure ladder access shall be provided;
  - (i) Wooden Scaffold not allowed
  - (j) Worker shall not be allowed to work from scaffolds during storms or high winds. After heavy rains or storm, the scaffolds should be inspected by the site-in-charge. Scaffolds should also be inspected every fortnight, during use, by him and again before starting use. Where the joints or members are found defective,

the joint should be set right and member replaced.

- (k) In case both light and heavy duty scaffolds are used in close vicinity, conspicuously placed notice boards shall indicate the light duty scaffolds and the limits on their usages.
- (l) Safety codes should be strictly followed. Standard safety requirements are just as important for temporary structures as they are for permanent structures

16.2.9 A team comprising of contractor's construction Manager and Safety Manager shall conduct a monthly working at height Safety Audit and the report shall be submitted no later than the 7<sup>th</sup> of each month within the Contractor's monthly OH&S Report.

### 16.3 Use of Ladders

16.3.1 All ladders shall be of sound construction and shall be free from patent defect. Ladder should meet the IS standard requirements

16.3.2 The Contractor shall implement a step platform/ladder inspection procedure requiring an identification method displaying company name, unique number, inspection frequency and inspection status.

16.3.3 Ladders should be checked daily and defective ladders shall be promptly and properly replaced.

16.3.4 Ladders shall not be used as working platforms.

16.3.5 Ladders shall;

- (a) be secured at the top or footed at the bottom to prevent slippage;
- (b) not be used if any rung is missing;
- (c) not be used for any other purpose than to provide access;
- (d) be set at an angle of seventy five degrees unless designed for vertical access;
- (e) all vertical ladders shall be fitted with hoops to prevent falls;
- (f) Wooden Ladder not allowed

### 16.4 Safety Harnesses / Fall Arresters

16.4.1. Where it is not possible to provide a safe working platform then the use of full body safety harness with double lanyard with fall arrester may be considered. If safety harness are used they should be of the full body type and secure anchorage points shall be provided and used. Workers must be instructed in the proper use of harnesses.

16.4.2. The Safety Harnesses/Fall Arresters should have valid Third-Party Examination Certificate by an approved competent person for every 6 months, before utilizing at site. Daily monitoring and checklist to be maintained for safety harness before commencement of specific work.

16.4.3 The Safety Harnesses/Fall Arresters should have valid Third-Party Examination Certificate by an approved competent person for every 6 months, before utilizing at site. Daily monitoring and checklist to be maintained before commencement of specific work.

## 17. Excavations

### 17.1 General

- 17.1.1 The Contractor shall appoint sufficient number of banksman to coordinate excavation activities at the pit and the haulage activities from the pit to the bank. The banksman shall be appointed in writing and should have attended Signalman training course approved by Employer/ Engineer. The banksman shall be properly identified on site, stationed at-grade and have overall control of the excavation works.
- 17.1.2 For excavations exceeding 4 metres in depth, the Contractor shall appoint one banksman. If deemed necessary by the Employer /Engineer, the Contractor shall appoint additional banksman. For excavations exceeding 10 metres in depth, the Contractor shall appoint one banksman for every long-arm or telescopic excavator at the bank.
- 17.1.3 Proper means of communication in the form of walkie-talkie sets should be established between the banksman and the excavator operators. No one shall be within any excavator's swing radius. In addition, all excavators shall be installed with rear view camera that enables the operator to have a clear view of the back of the machine
- 17.1.4 Long arm excavators shall be provided with an extended reflective mirror in front of the operator's cabin to enhance operator's visibility and shall not operate without the presence of the banksman.
- 17.1.5 Excavators within the excavation pit shall have suitably reinforced cabin roofs capable of withstanding impact from falling objects from the top of the excavation and its movement coordinated by one of its operator, who shall be appointed as a leader by the Contractor.
- 17.1.6 The designated locations at walers and struts used by the instrumentation contractor for instrumentation reading and monitoring shall be provided with 2 rows of horizontal rigid guardrails to prevent persons falling from height. Openings within struts are to be covered. The vertical distance between the 2 rows of horizontal guardrail shall be not more than 600mm. Toe boards are to be provided accordingly. Rebar shall not be used as edge protection only standard pipes should be used. The guard rail pipes shall be painted in reflective paints.
- 17.1.7 An alternative source of power and emergency lighting system shall be provided to allow emergency securing operations and evacuation safely in the event of a primary power failure. An adequate number of lamps shall be located at key points underground
- 17.1.8 Excavation is one of the important phases of any construction activity. Due to insufficient attention to the safety aspects it frequently becomes the cause of many accidents. Contractors are therefore required to plan and execute all excavations in a safe manner.
- 17.1.9 The contractor shall ensure that all excavations are supervised by workers with

thorough knowledge and experience of excavation work.

- 17.1.10 The integrity of the excavation and the support system shall be inspected prior to the commencement of any works on a daily basis with the results of the inspections being formally recorded. All such records shall be kept available for inspection by the Employer/ Engineer.
- 17.1.11 Where there is the possibility of any ingress of water then pumping sumps shall be established with pumps being readily available for use and additional ladders placed for use in the event of an emergency evacuation.
- 17.1.12 The contractor should not store, or move, materials and equipment near the edge of an excavation. Spoil and waste heaps should similarly be kept well one meter away from the edges of excavations.
- 17.1.13 Adequate and well-anchored stop blocks should be provided on the surface to prevent vehicles being driven into the excavation while tipping. The blocks should be placed at a sufficient distance away from the edge of the excavation to avoid the danger of it breaking away under the weight of the vehicles.
- 17.1.14 Make sure that there are safe means of access and egress, such as a properly secured ladder, if you are working in an excavation. This is of particular importance when there is a risk of flooding and rapid escape is essential.
- 17.1.15 There should be adequate lighting around the area of an excavation, particularly at access points and openings in barriers.
- 17.1.16 Shoring should be erected, altered or dismantled only by a competent worker operating under supervision. Wherever practicable, it should be installed before excavating to the final depth of the trench – it is necessary to begin when the trench is less than 1.2 m deep. The excavation and installation of shoring should then proceed by stages until the full depth is reached. You should be fully aware of the procedures to follow to rescue a fellow worker trapped by a fall of earth.
- 17.1.17 A team comprising of contractor's Construction manager and Safety Manager shall conduct a monthly Excavation Audit and the report shall be submitted no later than the 7<sup>th</sup> of each month within the Contractor's monthly OH&S Report.
- 17.1.18 Shoring protection and access shall be provided to all excavations, if the excavation depth is more than 1.5m

## **17.2 Planning**

- 17.2.1 The correct planning of excavations is essential for occupational Health and Safety and before digging any excavations Contractors should plan against the following;
- (a) Collapse of the sides;
  - (b) Materials falling onto people working in the excavation;
  - (c) People and vehicles falling into the excavation;
  - (d) People being struck by plant;
  - (e) Undermining nearby structures;

- (f) Contact with underground services.
- (g) Fumes; and
- (h) Make sure the necessary equipment needed such as trench sheets, props, etc., are available on site before work starts.

### 17.3 General Precautions

17.3.1 The following precautions should be observed;

- (a) Prevent the sides and the ends from collapsing by battering them to a safe angle or supporting them with timber, sheeting or proprietary support systems.
- (b) Do not go into unsupported excavations.
- (c) Never work ahead of the support.
- (d) Remember that even work in shallow trenches can be dangerous. You may need to provide support if the work involves bending or kneeling in the trench.
- (e) Prevention of materials falling into excavations
- (f) Do not store spoil or other materials within one metre of the sides of excavations. The spoil may fall into the excavation and the extra loading will make the sides more prone to collapse.
- (g) Make sure the edges of the excavation are protected against falling materials. Provide toe boards where necessary.
- (h) Wear a hard hat when working in excavations.
- (i) Take steps to prevent people falling into excavations. If the excavation is 1.5 m or more deep, provide substantial barriers, e.g. guard rails and toe boards.
- (j) Keep vehicles away from excavations wherever possible. Use brightly painted baulks or barriers where necessary.
- (k) Where vehicles have to tip materials into excavations, use stop blocks to prevent them from over-running. Remember that the sides of the excavation may need extra support.

### 17.4 Undermining nearby structures

17.4.1 The following precautions should be taken to prevent the undermining of nearby structures;

- (a) Make sure excavations do not affect the footings of scaffolds or the foundations of nearby structures. Walls may have very shallow foundations, which can be undermined by even small trenches.
- (b) Decide if the structure needs temporary support before digging starts. Surveys of the foundations and the advice of a structural engineer may be needed.

17.5 Avoiding underground services

17.5.1 The following precautions should be taken to avoid underground services;

- (a) Look around for obvious signs of underground services, e.g. valve covers or



patching of the road surface.

- (b) Use locators to trace any services by utility detector kit. Mark the ground accordingly.
- (c) Make sure that the person supervising excavation work has service plans and knows how to use them. Everyone carrying out the work should know about safe digging practices and emergency procedures.
- (d) Operate a “Permit to Dig” system.
- (e) Safety practices in sewage and septage management issued by the Chennai metropolitan water supply and sewerage board guidelines shall be followed.

## **18. Lifting Operations**

### **18.1 Lifting Appliances:**

- 18.1.1 The Contractor shall ensure that all lifting appliances, including synchronized mobile jacks, pit jacks, mobile cranes, tower cranes, gantry cranes, launching beams and lorry mounted cranes, prior to being allowed to work on site shall have available for inspection by the Employer / Engineer a current Certificate of Inspection issued by a Competent Person approved by CMRL. The Profile of the Inspection Agency, ISO certifications, inspection procedure, the name, educational qualifications and the competent person certificate issued by the Govt of Tamil Nadu are to be submitted to the Employer/Engineer a minimum 28 days before prior to appointment of the agency. Only upon notice of no objection by the Employer /Engineer shall be authorized to work on CMRL site. The approved competent person is authorized to conduct inspection and issue certificate for lifting machines, lifting appliances and lifting gears for two CMRL contract packages only.
- 18.1.2 Effective control must be exercised at all stages of a lifting operation through the deployment of a competent lifting supervisor, a qualified signaller, a registered crane operator and sufficient trained rigger.
- 18.1.3 The contractor shall ensure at a construction site of a building or other construction work that a competent person issues a certificate for the purpose of test and periodical examination of lifting appliances, loading of lifting appliances and lifting gears, ropes and heat treatment of lifting gears only after actual testing. All lifting machines, lifting appliances and lifting gears used on site must be suitable for the task, used within their rated safe load capacity and must be in good repair.
- 18.1.4 The Contractor is to submit a weekly list to the Employer /Engineer of all cranes permanently deployed on site detailing their ownership, make, identification number, maximum safe working load and the corresponding radius, lifting Machine (LM) certificate number and date of expiry.
- 18.1.5 All lorry loaders with articulating arm entering into the Site shall have an interlocking system installed to prevent a lorry loader with articulating arm from moving off before its articulated boom is fully retracted back to its “stored” position.



- 18.1.6 The boom of any lorry loader with articulating arm shall not be used for holding down materials or equipment and it shall be retracted to its “stored” position before it moves off.
- 18.1.7 Loading/ Unloading operations involving lorry loader with articulating arm shall take place over the side of the main body and not in an arc over either the front or rear of the chassis, in accordance with safe working practices.
- 18.1.8 No excavator shall be used as a lifting equipment and No Sheet piling activity by the excavator.
- 18.1.9 The Contractor shall ensure that a Permit to Lift system is operated to evaluate all routine planned lifting operations generically and all heavy or one-off lifting operations individually. This latter category of lifting operations includes the raising / lowering of items of plant in excavations. Lifting plans shall be submitted to the Employer /Engineer for acceptance in advance of any heavy or one-off lifting operations.
- 18.1.10 All lifting appliances with a lifting capacity of more than one tonne shall, where practicable, be fitted with Automatic Safe Load Indicators, Anemometer and Audible Warning Devices which shall be kept in an operable condition at all times the lifting appliance is in use. Checks should be made to ensure that the Automatic Safe Load Indicator and Anemometer is properly calibrated and is functioning properly. The Profile of the calibration Agency to be submitted to the Employer/Engineer a minimum 28 days before prior to appointment of the agency. Only upon notice of no objection by the Employer /Engineer shall be authorized to work on CMRL site.
- 18.1.11 The crane operator is to ensure that the outriggers are fully extended, and any adjustments made to the jacks to level the crane before the lifting operation commences. All lifting machine with outriggers, including lorry loader with articulating arm, shall have steel plates of minimum dimension 1m by 1m by 25mm placed under all the outriggers deployed for a lifting operation unless that crane is entirely sited on hard standing such as a reinforced concrete surface, with no void underneath. Pieces of timber are not to be used.
- 18.1.12 The Contractor shall ensure that no lifting operation shall be carried out on site using the auxiliary hook of a mobile crane unless the SWL of this is shown on the Lifting Machinery Certificate in addition to that of the main hook block, and is not exceeded.
- 18.1.13 The Contractor shall ensure that both the Lifting Supervisor and the crane operator are able to understand the working load chart.
- 18.1.14 Cranes fitted with a Automatic safe load indicator (ASLI) shall sound an audible alarm in the crane cab if its safe working load is exceeded on either the main or the auxiliary hook. A second alarm connected to the ASLI, shall be fitted external to the cab and shall emit a signal of a sufficient volume to make it audible above the ambient site noise levels during working hours. Visual warning shall also be provided externally to indicate safe working range and overload conditions. ASLI shall be connected directly to the start button and the ASLI should be enclosed with Plastic box or steel box.

- 18.1.15 load radius indicator shall be fitted with a limiting device, which disables the crane from continuing with any lifting operation under overload conditions. Once disabled, the device should only permit the crane to return to the safe working range. The device shall be tamper proof, with no over-ride to disable it, and shall be maintained in good working order. In case of provision of bypass switch for load radius indicator, it shall be secured by a lock and kept with the lifting supervisor.
- 18.1.16 The Contractor shall ensure that a sufficient number of qualified lifting supervisor are employed on site, whether by himself or by his Subcontractors, to give adequate cover for all lifting operations carried out both by day and by night including loading/unloading. There must be one lifting supervisor present for one crane including all minor and major operations. In addition the contractor shall also station at least one additional worker to assist signaller for lifting operations carried out near public areas.
- 18.1.17 The contractor shall install warning devices/ flags at least 2 metres above top of hoarding that are alongside roads, footpaths and adjacent structure.
- 18.1.18 lifting supervisor shall supervise and co-ordinate all lifting operations under his/her charge. lifting supervisor shall familiarize himself with the safe working load chart of each crane for which he is responsible and has identified himself to the operator of each such crane as being the only individual whose instructions concerning any lifting operation are to be followed. This includes the positioning of the crane prior to the lift as well as the slinging of the load.
- 18.1.19 Before any lifting operation involving a mobile or crawler mounted crane is carried out the lifting supervisor must satisfy himself that the crane is positioned suitably close to the load and its destination to ensure that the operation can be carried out at the safest appropriate radius.
- 18.1.20 The lifting supervisor shall ensure that the load is safely rigged, and a tag line is attached if appropriate, before signalling to the crane operator to start the lift. The lifting supervisor is responsible for the load until it is safely resting at the intended destination either by taking control of the operation himself for non-routine lifts or, for routine lifting operations, by thoroughly briefing the crane operator, riggers and signaller on the safe procedure to be followed.
- 18.1.21 The Contractor shall put measures in place to discipline any person other than the dedicated lifting supervisor, or one acting under his close supervision, who attempts to take control of any lifting operation other than those of a routine nature where a safe lifting procedure has already been established.
- 18.1.22 The Contractor shall engage only qualified crane operators with at least 5 years of experience in operating similar types of cranes with no record of crane toppling / failure or barred from any site previously for a crane related incident.
- 18.1.23 The crane operator shall enter the date, types of maintenance carried out and any malfunction of the crane in a checklist or logbook. He shall not operate the crane until any such defect is rectified, and the crane's use is authorized by his lifting supervisor.

- 18.1.24 All cranes shall be checked by its operator at the start of any day / shift using a checklist written in English and in a language comprehensible to the operator. Copies of the checklist together with LM certificate, operator certificate, permits should be retained in the crane cabin for verification.
- 18.1.25 Automatic safe load indicator (ASLI) shall be calibrated six-monthly by an approved agency and verified by the competent person during his six-monthly inspections. Records of load radius indicator calibration shall be submitted to the Employer /Engineer. If there is any doubt as to the accuracy of the load radius indicator the machine shall be taken out of use until the calibration is carried out to the satisfaction of the Employer /Engineer.
- 18.1.26 If the crane is down rated by the competent person during his six monthly inspections then the load radius indicator shall also be calibrated and a new capacity chart should be drawn up and posted in the crane cab. The competent person shall highlight the above details on the Lifting Machinery certificate.
- 18.1.27 Lifting machine shall not be used for any unsafe operation that may affect its overall integrity or stability.
- 18.1.28 The Contractor shall ensure that every lifting machine capable of travelling / tracking, including a gantry crane, or a slewing motion, such as a mobile crane, shall be operated in such a manner that there is always an unobstructed passageway in excess of 600mm between it and any fixture or other machinery.
- 18.1.29 All Lifting machines that operate on CMRL sites shall be installed with a rear view camera that enables the operator to have a clear view of the back of the machine.
- 18.1.30 The use of fly jib of any Lifting machines shall be subjected to the approval of the Employer /Engineer.
- 18.1.31 The concurrent use of the crane's main and auxiliary hooks during lifting shall be prohibited.
- 18.1.32 Before any crane is accepted onto site for use it must undergo a thorough mechanical check by the plant Manager, and the findings verified by the Contractor's Safety manager.
- 18.1.33 Lifting of machineries and/or equipment's shall be done as per manufacturer's recommendations and shall be subject to a regular preventative maintenance programme.
- 18.1.34 All lifting appliances shall be thoroughly tested and examined by a competent person approved by CMRL once at least in every six months or after it has undergone any alterations or repairs liable to affect its strength or stability. Within the validity, if the lifting appliances are shifted to a new site, re-examination by the same competent person for ensuring its safety shall also be done. Certificates of Inspection shall be available with the lifting appliance and a copy shall also be sent to the Employer/ Engineer.
- 18.1.35 The operators of lifting appliances shall conduct daily inspections of their respective

lifting appliances with the results of the inspections being recorded in the register and kept available for inspection by the Employer/ Engineer.

18.1.36 The Contractor shall ensure that only thoroughly third party agency trained and experienced persons aged twenty-one years and over are allowed to operate lifting appliances. The refresher training for Operator's shall be conducted every six months. The Profile of the training Agency, ISO certifications, course details, the name, educational qualifications and experience are to be submitted to the Employer/Engineer a minimum 28 days before prior to appointment of the agency. Only upon notice of no objection by the Employer /Engineer shall be authorized to work on CMRL site.

18.1.37 First generation hydra banned and shall not be strictly permitted to use in CMRL project sites.

18.1.38 All lifting machines, lifting appliances and lifting gears shall meet the requirements of Indian standards(IS).

18.1.39 The following Crane safety Equipment shall be install in the equipment

- a. As stipulated in Rule 57 of TBOCW Rules 2006, every lifting appliances and gears like cranes, hydras etc, if so constructed that the safe working load may be varied by raising or lowering of the jib or otherwise shall be attached with an automatic indicator of safe working loads approved by Bureau of Indian standards/ International certifying bodies which gives a warning to the operator and arrests further movements of the lifting parts.
- b. ASLI Interlocked with the equipment key and Bye pass button shall be replaced with Key and the key shall be available with plant department. Log register should maintain for the purpose of the key is used.
- c. ASLI shall be covered with plastic or steel cover.
- d. Motion limit Devices
- e. Overload cut out devices
- f. Level indicator
- g. Anemometer
- h. Machinery guarding
- i. Proximity sensor
- j. Visual Warning and alarm
- k. Sensing devices
- l. Boom angle indicator

18.1.40 The special assessment as per IS 14475-1(1997) and ISO 12482-1:1995 shall be carried out later than the following number of years after manufacture for:

- Tower cranes, loader cranes, mobile cranes: 10 years;
- All other cranes: 20 years.

18.1.41 Lifts involving two or more cranes are complex operations requiring considerable skill

and planning. As a result, multiple crane lifts (or tandem lifts) must be planned and carried out under the supervision of a competent person.

18.1.42 The exact weight each crane will carry must be determined in advance since all cranes must operate in unison. Operator headset communications systems should be implemented. For these more complicated lifts, no more than 75 percent of the net capacity of any crane should be used. Because the load weight shifts from one crane to the other during movement of the booms, these changes need to be considered during the planning process. Pre-planned load considerations can also be disrupted if one crane lifts its end of the load before the other crane lifts, causing the first crane to be overloaded. A practice run without the load is recommended to help coordinate the process.

18.1.43 Compliance with the crane manufacturer's recommended wind-velocity shutdown limit is required for safe operations, though many operators halt work if the wind speed reaches a generic 30 mph.

18.1.44 All lifting appliances including all parts and gears thereof whether fixed or movable, are tested and examined by a competent person before being taken into use for the first time or after it has undergone any alterations or repairs liable to affect its strength or stability or after erection on a construction site and also once at least in every five years, in the manner specified in Schedule-I of TBOCWR 2006.

18.1.45 Certificate in respect of the following is obtained from competent person in the forms as per

the TBOCWR 2006 under schedule XIII mentioned below, namely: -

- Winches, derricks and their accessory gears in Form V.
- Cranes or hoists and their accessory gears in Form VI.
- In case of test, examination and re-examination of loose gears Form VII.
- In case of test and examination of wire ropes Form VIII.
- In case of heat treatment and examination of loose gears Form IX.

18.1.46 All alarms and signals like automatic safe load indicators (ASLI), boom angle indicators, boom extension indicators, over lift boom alarm, swing alarm, hydraulic safety valves, mechanical radius indicators, load moment indicators etc. shall be examined daily basis by the operator. All devices shall be maintained in good working condition.

## **18.2 Lifting Gear:**

18.2.1 Lifting Gear includes chain slings, rope slings, or similar gear and a ring, link, hook, plate clamp, shackle, swivel or eye bolt.

18.2.2 The Contractor shall ensure that all lifting gear shall be in good condition and shall be tested and certified every six months, with the Safe Working Load being stamped or clearly displayed upon it. Records of test shall be kept available for inspection by the

Employer/ Engineer.

- 18.2.3 The Contractor shall ensure that Lifting Gear is not loaded beyond its SWL and this includes multi leg chain slings being used at variable angles.
- 18.2.4 The Contractor shall ensure that Lifting Gear is not used for any purpose other than the raising or lowering of a load. If an excluded activity takes place, such as the use of a lifting chain for towing an item of plant / machinery, then the Contractor shall ensure that this item of Lifting Gear is no longer used for any further lifting operations.
- 18.2.5 The Contractor shall implement an inspection programme to thoroughly check all Lifting Gear by a lifting supervisor prior to its first use on site and thereafter on a monthly basis. A monthly colour coding system shall be adopted. Defective Lifting Gear shall be discarded.
- 18.2.6 All lifting gear shall be visually inspected before any use and if any defects are found then it shall be removed from site or dismantled / disabled in order to ensure that it is not used in a defective state.
- 18.2.7 All lifting gear shall be properly stored and not left lying on the ground where it could be damaged or used in an unsafe manner.
- 18.2.8 A lifting gear is initially tested for the manufacturer by a competent person, in a manner specified in Schedule-1 of TBOCWR 2006 before taking into use or after undergoing any substantive alterations which renders its any part liable to affect its safety and such gear altered shall subsequently be re-tested for the use of its owner at least once in every live years. Certificates of initial and periodical tests and examination of loose gears under TBOCWR 2006 are obtained in Form VII under Schedule – XIII of TBOCWR 2006.

### **18.3 Lifting Operations:**

18.3.1 The Contractor shall ensure that during the course of any lifting operations the following minimum requirements shall be followed:

- (a) All lifting operations shall be under the control of a competent “Lifting Supervisor” appointed by the contractor.
- (b) Only thoroughly third party agency trained and experienced crane drivers shall be allowed to operate cranes. The refresher training for operator’s, lifting supervisor, banks Man and rigger etc. shall be conducted every six months.
- (c) Only thoroughly third party agency trained and experienced slingers and riggers shall be allowed to sling loads and give directions to crane operators.
- (d) A standard code of hand signals shall be adopted for controlling the movements of the crane and both the driver and the signaller shall be thoroughly familiar with the signals.
- (e) The driver of the crane shall respond to signals from only the appointed signaller but shall obey the stop signal at any time no matter who gives it.
- (f) Before commencing any lifting operations, the ground conditions on which the



crane is to stand shall be investigated in order to ensure that the load bearing capabilities are adequate.

- (g) The weight of the load must be known to the crane driver and the slinger/rigger before lifting commences.
- (h) No loads are to be slewed over public areas without stopping pedestrians and vehicles first.
- (i) No unauthorised persons are allowed into the lifting zone.
- (j) No person is allowed to ride the hook of the crane or the loads being lifted.
- (k) Any areas where a minimum clearance of six hundred millimetres from the rear of the slewing kentledge of the crane cannot be achieved and where persons could be trapped against obstacles then a fence shall be erected to prevent access.
- (l) All crane hooks shall be fitted with an operable Safety Latch.
- (m) Wherever practicable all loads shall have tag-lines attached in order to ensure that the load can be controlled at all times.
- (n) Provision shall be made to ensure that the lifting slings or chains can be safely removed from the loads once they have been landed.
- (o) All lifted loads and stacked materials shall be left in a secure and stable condition at all times.
- (p) Whenever working close to isolated overhead power-lines the lifting appliances shall be grounded to earth as a secondary precaution against accidental energisation.
- (q) No close working to any live overhead power-lines is permitted without the operation of a strict Permit to Work system being in place.
- (r) Install wireless anti-collision sensory protection device and alarm system before tandem lifting operation commences.
- (s) Operators, lifting supervisor and Riggers is medically examined periodically as specified in Schedule-VII of TBOCWR 2006.

## 19. Work in Confined Spaces

### 19.1 General

19.1.1 The term 'confined space' has three defining features as , any space:

- (a) in which a person can bodily enter;
- (b) which has limited entry / egress; and
- (c) which is not meant for occupancy.

Generally it is place which is substantially (though not always entirely) enclosed with a reasonably foreseeable risk of serious injury from hazardous substances or conditions within the space or nearby.

19.1.2 Some confined spaces are fairly easy to identify, for example, closed tanks and sewers.



Others are less obvious but may be equally dangerous, for example closed and unventilated or inadequately ventilated rooms and silos, ducts, culverts, tunnels, boreholes, bored piles, manholes, shafts, excavations, sumps, inspection pits, cofferdams, and building voids.

## **19.2 The hazards**

19.2.1 The most likely hazards are as follows:

- (a) Flammable Substances and Oxygen Enrichment;
- (b) Toxic Gas, Fume or Vapour;
- (c) Oxygen deficiency;
- (d) The Ingress or Presence of Liquids;
- (e) Presence of Excessive Heat,
- (f) Excessive Humidity.

## **19.3 Entry Procedures**

19.3.1 Contractors will ensure that no work will be undertaken in Confined Spaces unless a Permit to Work, has been prepared and issued.

19.3.2 Only persons who have been thoroughly third party trained, experienced and are physically fit shall be allowed to work in Confined Spaces. The refresher training shall be conducted every six months. The Profile of the training Agency, ISO certifications, course details, the name, educational qualifications and training experience for of the trainers are to be submitted to the Employer/Engineer a minimum 28 days before prior to appointment of the agency. Only upon notice of no objection by the Employer /Engineer shall be authorised to deliver training on CMRL sites.

19.3.3 Persons with any of the following medical conditions shall not be allowed to work in confined spaces:

- (a) A history of fits, blackouts or fainting attacks,
- (b) A history of heart disease or disorder,
- (c) High blood pressure,
- (d) Asthma bronchitis, or shortness of breath on exertion,
- (e) Deafness
- (f) Meniers disease or disease involving giddiness or loss of balance,
- (g) Claustrophobia or nervous or mental disorder,
- (h) Back pain or joint trouble that would limit mobility in confined spaces,
- (i) Deformity or disease of the lower limbs limiting movement.
- (j) Chronic skin disease,
- (k) Serious defects in eye sight or lack of sense of smell

19.3.4 Before any confined space work commences the following equipment shall be available for use and the sample of the personal protective equipment to be submitted to the Employer's /Engineer for notice for no objection. Only upon the notice of no objection

by the Employer /Engineer shall be authorised to use on CMRL site.

- (a) Multi Gas Monitor; or other suitable gas monitoring equipment.
- (b) Minimum 35 numbers of 60 minutes Self Contained Breathing Apparatus (Approved by DGMS) at each Tunnel entrance shall available and Minimum 6 numbers four hours closed circuit breaker breathing apparatus (CCBA Approved by DGMS) shall available for each tunnel to enable rescue to be carried out;
- (c) Full Body Type Harness for each worker;
- (d) Tripod and Lifeline Hoist Rope; for work in situations where a vertical exit from the confined space is required.
- (e) Flame-proof lighting. (Hand lamps not more than 24 volts.);
- (f) Resuscitation Equipment and details of Emergency control centre and operating Procedure to be submitted to the Employer's /Engineer for notice for no objection
- (g) Ventilation Equipment.

The persons involved in the confined space working operations shall need to be thoroughly trained and certified as being competent in the use of the above detailed item of equipment.

- 19.3.5 The Contractor shall have controlled access/egress points to confined spaces to prevent unauthorised access. The Contractor shall ensure that there are at least two readily accessible escape routes from each confined space.
- 19.3.6 The Contractor shall operate a tag system for entry so that all personnel entering the confined space can be accounted for.
- 19.3.7 The Contractor shall ensure that there is an certified man-riding cage capable of taking a stretcher and two persons, together with an identified crane equipped with rescue equipment, on standby at all times whilst work is carried out in the confined space.
- 19.3.8 Gas monitoring shall be conducted by a competent confined space assessor to certify that the confined space is safe for workers to enter and thereafter at every four (4) hours intervals.
- 19.3.9 In addition, the Contractor shall ensure that suitable atmospheric monitoring devices such as anemometer and wet/dry bulb thermometer are made available for the competent confined space assessor to determine the air flow, ambient temperature and humidity level within the confined space.
- 19.3.10 Provide noise enclosure for the ventilation fan and install silencer at the end of the ventilation fan.

## 20. Site Electricity

## 20.1 General

- 20.1.1 The Contractor shall nominate a representative whose name and qualifications shall be submitted in writing to the Employer/ Engineer for review not later than 4 weeks before the appointment and who shall be solely responsible for ensuring the Safety of all temporary electrical equipment on Site. The Contractor shall not install or operate any temporary Site electrical systems until this representative is appointed and has commenced duties. The project specific CEIG shall be obtained by the contractor as per Central Electrical Authority (Safety Requirements for construction, operation and maintenance of Electrical plants and electric lines) Regulations, 2011.
- 20.1.2 The name and contact telephone number of the electrical C licence holder shall be displayed at the main distribution board for the temporary electrical supply so that he can be contacted in case of an emergency.
- 20.1.3 The Contractor shall submit schematic diagrams and the details of the equipment for all temporary electrical installations, and these diagrams together with the temporary electrical equipment shall be submitted to the Employer/ Engineer for review.
- 20.1.4 All electrical installation work on Site shall be carried out in accordance with the requirements laid down in the Specification. All work shall be supervised or executed by an Govt recognized Electrical ‘C’ licence holder.
- 20.1.5 All temporary electrical installations, equipment and tools shall be checked and certified safe for use prior to usage on site by a full-time an Govt recognized Electrical ‘C’ licence holder and thereafter monthly and after any repairs. The Electrical ‘C’ licence holder shall provide a sticker on the equipment and tools indicating the date of inspection and is safe for use.
- 20.1.6 A current photograph of the Electrical ‘C’ licence holder’s and the contact number(s) shall be displayed on the outside of all boxes containing electrical DBs for ease of reference. These boxes must be secured with locks to prevent tampering and the keys kept with the Electrical ‘C’ licence holder /Contractor’s Safety Dept.
- 20.1.7 The Contractor shall ensure that all portable electrical appliances used above and below ground level, that are hand held tools and inspection lamps, are rated at 110 volts AC via a step down transformer centre tapped to earth (CTE).
- 20.1.8 The Contractor shall ensure that all generators and welding sets in use on Site are adequately and effectively earthed at all times during operation.
- 20.1.9 All Temporary Electrical Site installations and distribution systems shall as a minimum meet IP44/IP 65 standards and be in accordance with:
- (a) Indian Electrical Regulations;
  - (b) The Power Companies’ Supply Rules;
  - (c) BS 7671 Requirements for electrical installation, the IEE Wiring Regulations (16th Edition);
  - (d) BS 7375 Distribution of Electricity on Construction and Building Sites;

- (e) BS 4363 Distribution Assemblies for Electricity Supplies for Construction and Building Sites; and
- (f) BS 6164 Safety in Tunnelling in the Construction Industry.

## 20.2 Design Considerations

20.2.1 Distribution equipment utilised within the temporary electrical distribution system shall incorporate the following features:-

- (a) Flexibility in application for repeated use;
- (b) Suitability for transport and storage;
- (c) Robust construction to resist moisture and damage; and
- (d) Safety in use.

20.2.2 All cabling shall be run at high level whenever possible and firmly secured to ensure it does not present a hazard or obstruction to people and equipment.

20.2.3 The installation on Site shall allow convenient access to authorised and competent operatives to work on the apparatus contained within.

## 20.3 Distribution of supply

20.3.1 The Site mains voltage shall be as the Electricity Utility supplies, 415V 3-phase 4-wire system.

- (a) Single-phase voltage shall be as the Electricity Utility supplies, 240V supply.
- (b) Reduced voltages shall conform to BS 7375.

20.3.2 The following voltages shall be adhered to for typical applications throughout the distribution systems:

- (a) Fixed plant - 415V 3 phase.
- (b) Movable plant fed by trailing cable - 415V 3 phase.
- (c) Installations in Site buildings - 240V 1 phase.
- (d) Fixed flood lighting - 240V 1 phase.
- (e) Portable and hand held tools - 110V 1 phase.
- (f) Site lighting (other than flood lighting) - 110V 1 phase. and
- (g) Portable hand-lamps (general use) - 110V 1 phase.

20.3.3 Protection shall be provided for all main and sub-circuits against excess current, residual current and earth faults. The protective devices shall be capable of interrupting (without damage to any equipment or the mains or sub-circuits) any short circuit current that may occur.

20.3.4 Earthing and bonding shall be provided for all electrical installations and equipment to prevent the possibility of dangerous voltage rises and to ensure that faults are rapidly cleared by installed circuit protection.

20.3.5 Only plugs and fittings of the weatherproof type shall be used and they should be colour

coded in accordance with the Internationally recognised standards for example as detailed as follows:

- (a) 110 volts: Yellow.
- (b) 240 volts: Blue.
- (c) 415 volts: Red.

## 20.4 Cables

20.4.1 Cables shall be selected after full consideration of the conditions to which they will be exposed and the duties for which they are required. For supply cables up to 3.3kV the cable armouring shall be used as the earth return in conditions where the cable is continuously extended and not subject to continuous movement after installation. If cable laid in the above ground or underground should be laid as per the norms of Indian electricity rules 1956.

20.4.2 For supplies to mobile or transportable equipment where operation of the equipment subjects the cable to flexing, the cable shall conform to one of the following specifications appropriate to the duties imposed on it:

- (a) BS 6708 flexible cables for use at mines and quarries;
- (b) BS 6007 rubber insulated cables for electric power and lighting; and
- (c) BS 6500 insulated flexible cords and cables.

## 20.5 Maintenance

- Strict maintenance and daily checks of control apparatus and wiring distribution systems shall be carried out by an Govt recognized Electrical ‘C’ licence holder (duly qualified to carry out the said checks) to ensure safe and efficient operation of the systems. A team comprising of contractor’s senior Electrical Manager and Safety Manager shall conduct a monthly Electrical Safety Audit and the report shall be submitted no later than the 7<sup>th</sup> of each month within the Contractor’s monthly OH&S Report.
- All portable electrical appliances shall be permanently numbered (inspection tag labels or similar) and a record kept of the date of issue, date of the last inspection carried out and the recommended inspection period.
- The Work area light mounting height was 25 ft (7.6 m) and the distance between two work area light should be within 30 meters. The Lighting pole shall not be tied on the barricading board.

## 20.6 Lighting:

20.6.1 The contractor shall provide sufficient site lighting, of the right type and at the right place for it to be properly effective. Lighting ought not to introduce the risk of electric shock. Therefore, 230V supplies should be used for those fittings, which are robustly installed, and well out of reach e.g. flood lighting or high-pressure discharge lamps.

## 20.7 Illumination

20.7.1 The contractor shall conduct a monthly illumination monitoring by lux meter for all the locations and the report shall be sent to the Employer/ Engineer within 7th of the next month and the same shall be reviewed during the monthly OSH&E committee meeting.

20.7.2 The contractor shall take every effort to illuminate the work site is listed in the below table

### Minimum Lighting Requirements

| S.N. | Facility or Function                                                                                                                                                                                                                                                            | Luminance – lx<br>(lm/ft <sup>2</sup> )                          |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| 1.   | Administrative areas (offices, drafting and meeting rooms,                                                                                                                                                                                                                      | 540 (50)                                                         |
| 2.   | <b>Construction areas</b><br><ul style="list-style-type: none"> <li>- general indoor</li> <li>- general outdoor</li> <li>- tunnel and general underground work areas<br/>(minimum 110 lux required at tunnel and shaft heading during drilling, mucking and scaling)</li> </ul> | 55 (5)<br>33 (3)<br>110<br>(10)                                  |
| 3.   | <b>Access ways</b><br><ul style="list-style-type: none"> <li>- exit ways, walkways, ladders, stairs</li> </ul>                                                                                                                                                                  | 110 (10)                                                         |
| 4.   | <b>Maintenance / Operating areas / shops</b><br><ul style="list-style-type: none"> <li>- vehicle maintenance shop</li> <li>- carpentry shop</li> <li>- outdoors field maintenance area</li> <li>- refuelling area, outdoors</li> <li>- shops, fine details work</li> </ul>      | 325 (30)<br>110 (10)<br>55 (5)<br>55 (5)<br>540 (50)<br>325 (30) |
| 5.   | Mechanical/electrical equipment rooms                                                                                                                                                                                                                                           | 110 (10)                                                         |
| 6.   | Hoists, Elevators, freight and passenger                                                                                                                                                                                                                                        | 215 (20)                                                         |
| 7.   | <b>Warehouses and storage rooms/area</b><br><ul style="list-style-type: none"> <li>- indoor stockroom, active/bulk storage</li> <li>- indoor rack storage</li> <li>- outdoor storage</li> </ul>                                                                                 | 110 (10)<br>270 (25)<br>33 (3)                                   |
| 8.   | Health Centres and First aid stations and infirmaries                                                                                                                                                                                                                           | 325 (30)                                                         |
| 9.   | Toilets, wash and dressing rooms                                                                                                                                                                                                                                                | 110 (10)                                                         |
| 10.  | Work areas – general (not listed above)                                                                                                                                                                                                                                         | 325 (30)                                                         |
| 11.  | Parking areas                                                                                                                                                                                                                                                                   | 33 (3)                                                           |
| 12.  | Visitor areas                                                                                                                                                                                                                                                                   | 215 (20)                                                         |
| 13.  | Laboratories                                                                                                                                                                                                                                                                    | 540(50)                                                          |



## **20.8 Monthly Electrical Safety Audit**

20.8.1 A team comprising of contractor's Senior OHS&E (Electrical) manager and the Employer/ Engineer shall conduct a monthly electrical safety audit covering the following and submit the report to the Employer/ Engineer.

- i) Electrical accidents investigation findings and remedy
- ii) Adequacy of power generation and power requirements
- iii) Power distribution and transmission system in place
- iv) Updated electrical single line diagram showing the current condition of power source and distribution including the IP44, IP65 DBs arrangement.
- iv) Electrical protection devices – selection, installation and maintenance.
- v) Earth or ground connection and earth pit maintenance details
- vi) Education and training of electrical personnel undertaken
- vii) Routine electrical inspection details
- ix) Electrical maintenance system and register.
- x) Name plate details of major electrical equipment
- xi) CEIG Approval
- xi) Classified zones in the site, if any.

## **21. Welding and Cutting**

### **21.1 General**

21.1.1 Contractors shall ensure that all welding, cutting and gouging is carried out so that the risks are kept at a minimum. There will be some circumstances when Permits to Work will need to be issued, such as

- (a) Working in tunnels;
- (b) Welding over areas where others are working;
- (c) Working in areas with increased fire risks or hazardous environments;

21.1.2 All equipment must be in good condition, properly installed and routinely inspected by a competent person, and records must be kept available for inspection by the Employer/ Engineer.

21.1.3 Flexible hoses, cables and connections must be free from damage or risk of damage in service. Cables and hoses shall have adequate carrying capacity.

21.1.4 Welders shall wear the correct personal protective equipment which includes the following;

- (a) Face and eye protection with correct grade of shield;
- (b) Gauntlet gloves;
- (c) Safety footwear
- (d) Welders apron or fire retardant overalls;
- (e) The atmosphere in the vicinity of work must be known to be safe to breathe and free from flammable gases.

- 21.1.5 Adequate ventilation and fume extraction must be provided and used as required by the risk assessment and especially in enclosed areas and pits.
- 21.1.6 Surfaces to be heated by the process must be cleaned of contaminants that may be degraded by heat or give off noxious fumes (e.g. paints, plastics, zinc coating).
- 21.1.7 Naked flames or high temperature surfaces must not be allowed in the vicinity of volatile solvents.
- 21.1.8 All moveable flammable materials must be removed from the vicinity of work and fireproof covers placed over all flammable materials that cannot be removed.
- 21.1.9 During all welding the work piece and any access equipment must be safely secured.

## **21.2 Oxy-fuel Gas Processes**

- 21.2.1 Handle cylinders carefully, keep outside enclosed areas and secure in an upright position on a custom-built stand or trolley fitted. The metal cap shall be kept in place to protect the valve when the cylinder is not connected for use.
- 21.2.2 Flash back arresters shall be fitted to both the fuel gas and oxygen cylinders;
- 21.2.3 Non return valves shall be fitted to the torch or cutting torch;
- 21.2.4 Hose clamp or clip shall be used to connect hoses firmly in both sides of cylinders and torches.
- 21.2.5 Close cylinder valves when flame is extinguished.
- 21.2.6 Ensure any vessel, drum or tank that has contained flammable or toxic substances has been properly cleaned and inspected before subjecting it to hot work.
- 21.2.7 Checks for gas leaks should be carried out using soapy water.
- 21.2.8 Remove all torches from enclosed areas when not in use.
- 21.2.9 6kg fire extinguisher to be available at all places where hot work is being carried out.
- 21.2.10 Use firewatchers if there is a possibility of ignition unobserved by the operator (e.g. on the other side of bulkheads).

## **21.3 Arc Cutting, Gouging and Welding Processes**

- 21.3.1 Connect the welding current return cable to the work piece close to the arc point or to a well electrically conductive support structure in good contact with the work piece. Also, connect the work piece or the support structure to a separate earth terminal.
- 21.3.2 Take precautions against the risk of increased fume hazards when welding with chrome containing fluxed consumables or high current metal inert gas (MIG) or tungsten inert gas (TIG) processes e.g. local exhaust ventilation.
- 21.3.3 Avoid being in contact with water or wet floors when welding. Use duckboards or rubber protection.
- 21.3.4 Provide screens to limit exposure of others to glare from arcs.
- 21.3.5 Use the correct eye and face protection with the correct filter glass.
- 21.3.6 Use a low voltage open circuit relay device if welding with alternating current in

constricted or damp places.

- 21.3.7 All electrical installations shall meet the IS: 5571: 1997 and NFPA 70 for gas cylinder storage area and other hazardous areas.
- 21.3.8 Transformer used for electrical arc welding shall be fixed with Ammeter and Voltmeter and also fixed with separate main power switch.
- 21.3.9 Welding grounds and returns should be securely attached to the work by cable lugs, by clamps in the case of stranded conductors, or by bolts for strip conductors. The ground cable will not be attached to equipment or existing installations or apparatus.
- 21.3.10 The current for Electric arc welding shall not exceed 300 A on a hand welding operation.
- 21.3.11 Use a low voltage open circuit relay device if welding with alternating current in constricted or damp places.

## **22. Compressed Gases**

### **22.1 Storage**

- 22.1.1 The Contractor shall ensure that all compressed gases, such as oxygen and fuel gases, are stored in a safe manner in keeping with the following requirements.
- 22.1.2 When not in use compressed gas cylinders should preferably be stored in the open air in a well-ventilated area at ground level on a firm level surface at least 3m away from any cellars, drains, excavations or other hollows where vapour may collect. There should be good access to the area, which should be kept clean and clear of combustible material, including wood, packing materials and vegetation. If any protection is provided to prevent cylinders being exposed to the weather, it should be of non-combustible material and should not inhibit ventilation. The area should not be close to any source of heat.
- 22.1.3 Oxygen cylinders and flammable gas cylinders shall be stored separately, at least 6.6 meters (20 feet) apart or separated by a fireproof, 1.6 meters (5 feet) high partition. Flammable substances shall not be stored within 50 feet of cylinder storage areas.
- 22.1.4 It is important that the valves of so-called 'empty' cylinders are kept closed as well as those of full cylinders and that plugs, shrouds and caps are kept in place on all cylinders. This is necessary not only to prevent the escape of any residual compressed gas into the atmosphere but also to ensure that air is not sucked into the cylinder to form an explosive mixture inside it. All cylinders should be stored with their valves uppermost.
- 22.1.5 The storage area should be enclosed by a fence approximately 2 metres in height. The fence should be made of non-combustible material and should not inhibit natural ventilation, particularly at low level - a wire mesh fence is particularly suitable for this purpose. The fence should have at least two means of exit, which should not be adjacent to each other. The gates should open outwards and not be self-locking. Both exits should be unlocked when persons are within the storage compound. At all times when the site is unattended the storage area should be secured.

- 22.1.6 On sites where only small quantities of compressed gas are stored (i.e. less than 300 kg). Only one exit will be necessary providing there is no risk of a person being trapped in the enclosure. The cage should be clearly marked "Highly Flammable and notices prohibiting smoking and naked lights should be displayed.
- 22.1.7 portable first aid and 6kg fire extinguishers shall be positioned in close proximity to the storage area for use in an emergency. Fire Extinguisher should confirm to IS 2190: 1992.
- 23 Before utilizing LPG at CMRL site premises, contractor should get permission from Employer/engineer. The storage of gas cylinders conforms to local/national regulation.

### **23.1 Handling Compressed Gas cylinders**

- 23.1.1 Cylinders should be handled with care and wherever practicable moved on specially designed trolleys. The valve on a cylinder should not be used for lifting or to lever the cylinder into position. Damage to the valve can result in highly dangerous situations following the escape of gas. For the same reason throwing or dropping cylinders should be prohibited as in such circumstances damage to the valve is even more likely.
- 23.1.2 Before connecting any cylinder or container of compressed gas to equipment it is essential that all fires, flames or other sources of ignition in the vicinity, including cigarettes and pilot lights, are extinguished. Where practicable cylinders should be changed in the open air. The cylinder should be examined and any damaged or faulty cylinder should NOT be used. No attempt should be made to rectify any fault or damage. The cylinder should be put in a safe place away from other cylinders or combustible materials until returned to the supplier.
- 23.1.3 If a cylinder is found to be leaking and the leak cannot be stopped, the cylinder should be carefully removed to a well-ventilated open space free from sources of ignition. It should be left with the leak, usually at the valve, uppermost, marked faulty and notices displayed prohibiting smoking or other naked lights. General access should be prevented by barriers or otherwise. The supplier of the cylinder should be informed immediately. Under no circumstances should users attempt to dismantle or repair defective cylinders.

### **22.3 Regulators**

- 22.3.1 Regulators should be suitable for the gas and pressure in use. Checks for leaks at the regulator nuts should be made only by using soapy water. In the event of a defect or of any damage to a regulator, no attempt should be made to repair it. Such repairs should only be carried out by specialists.

### **22.4 Hoses**

- 22.4.1 Flexible tubing should only be used for final connections to appliances. Flexible hoses should comply with BS 3212, BS 5120 or other nationally recognised standard. They should be additionally protected or of steel braid reinforced construction wherever they might be subject to damage by abrasion and so sited that they are not exposed to excessive heat. The length of hoses should be kept as short as practicable

## **22.5 Training and Instruction**

22.5.1 Many accidents involving compressed gas are due to ignorance of simple basic precautions. It is essential that all persons using compressed gas are suitably instructed about the hazards and the precautions to be taken in its use.

## **23. Machinery**

### **23.1 Machinery Fencing**

23.1.1 The Contractor shall ensure that all gears, revolving shafts, flywheels, couplings and other dangerous parts of machinery shall be effectively guarded unless they are so constructed, installed or placed as to be safe as if they were guarded.

23.1.2 Fencing of dangerous parts of machinery shall not be removed while the machinery is in use or in motion. If the fencing is required to be removed for maintenance purposes it shall be replaced before the machine is taken into use.

### **23.2 Maintenance**

23.2.1 The Contractor shall ensure that all machinery used on site is in safe condition and is properly maintained and repaired by duly authorised, thoroughly trained and experienced persons.

23.2.2 No repair to machinery shall be carried out whilst it is in motion unless it is unavoidable.

23.2.3 Maintenance records shall be kept available for inspection by the Engineer.

### **23.3 Air Receivers**

23.3.1 All Air receivers shall be fitted with a pressure relief valve and shall have the safe working pressure clearly marked upon them.

23.3.2 Every air receiver shall be subject to an annual test, which shall be carried out by a duly authorised person. The results of all tests shall be recorded and the records shall be kept available for inspection by the Employer/ Engineer.

23.3.3 The connection couplers on compressed airlines shall be securely fixed together and have Safety chains or be wired at the joints in order to ensure that the joints do not come apart when charged with compressed air.

### **23.4 Woodworking Machines**

23.4.1 All woodworking machines shall be fitted with the following guards and devices;

- (a) Top Guard;
- (b) Riving Knife;
- (c) Guards to protect all drive belts etc.;
- (d) An emergency stop switch easily accessible by the operator;
- (e) A push stick;

23.4.2 Woodworking machines shall be operated only by thoroughly trained and experienced operators.

### **23.5 Abrasive Wheels**

23.5.1 All Abrasive wheel machines shall be fitted with appropriate guards which shall be kept in place at all times the machine is in use.

23.5.2 All abrasive wheel machines shall have the spindle speed clearly marked upon them in revolutions per minute.

23.5.3 Only thoroughly trained and experienced persons are allowed to change the wheels on the machines. Wheels must inspected and ring tested before mounting to ensure that wheels are free from cracks or defects.

23.5.4 Safety Goggles or Face shields must be worn when grinding or cutting with abrasive wheels.

## 24.Plant and Equipment Operations

### 24.1 General

- 24.1.1 The contractor shall ensure that only safe and well-maintained plant and equipment shall be allowed to operate on any of the sites.
- 24.1.2 All operators of heavy plant such as, earth movers, piling rigs, etc. shall be medical Examination and Vision Test over twenty years of age and be thoroughly trained and experienced to operate the equipment. Vision test shall be carried out by the ophthalmologist doctor.
- 24.1.3 No unauthorized person shall be permitted to ride on plant.
- 24.1.4 The operators shall conduct daily inspections of their respective items of plant with the results of these inspections being recorded and the records kept available for inspection by the Engineer.
- 24.1.5 All mobile heavy plant shall be equipped with at least one 6 kg Dry Powder Fire Extinguisher, carried at a suitable position so as to ensure its easy availability.
- 24.1.6 Whenever heavy plant is operating in congested areas, thoroughly trained and experienced banksmen shall be deployed to control the plant and personnel movement and interface.
- 24.1.7 Any waste engine oil and filters following any on site servicing and maintenance shall be removed from the sites and disposed of in an environmentally conscious manner at authorised disposal locations.
- 24.1.8 All drums of fuel oil shall be stored on drip trays or the fuel shall be kept in banded bulk storage fuel tanks, with quantities stored being kept to a minimum.
- 24.1.9 The storage areas shall have dry powder fire extinguishers positioned in close proximity to their location for use in an emergency.
- 24.1.10 The Contractor shall assess the OSH&E risks especially in terms of age, noise, emissions, condition, etc. associated with the plant, equipment or tool and only those assessed with minimal OSH&E risks shall be brought to the Site.
- 24.1.11 The Employer /Engineer shall stop the plant, equipment or tool from operation or require its removal if he finds the OSH&E associated risks to be high. The Contractor shall not be entitled to any claim for compensation or Extension of Time for Completion.
- 24.1.12 The Contractor shall implement a preventive maintenance programme to ensure that all plant, equipment and tools are maintained in a safe and working order.
- 24.1.13 The Contractor shall implement a monthly inspection program to inspect all plant, equipment and tools. All plants, equipment and tools that have undergone repair or maintenance shall be inspected and checked before being returned to service. Stickers or tags shall be displayed to indicate its approval for usage or “Not for Use”.
- 24.1.14 The Contractor shall implement a lockout and tag-out system, Energy Lockout and



### Tagout Procedure.

24.1.15 Job-made or modified tools of any kind shall not be used on site.

24.1.16 All new heavy equipment to be used in CMRL Worksites shall be subjected to approval of the Employer /Engineer.

24.1.17 On-site speed limits shall be applied as a mandatory standard and due diligence shall be given to any operations which are likely to create a dust nuisance.

24.1.18 All machinery/plant/vehicle used on site shall have enclosed air-conditioned cabs.

24.1.19 The windows shall be provided with toughened safety glass and wipers.

24.1.20 Seat belt for the driver shall be provided.

24.1.21 All site plant and machinery shall be thoroughly washed down before leaving the site.

24.1.22 A team comprising of contractor's senior plant Manager and Safety Manager shall conduct a monthly plant and machinery Audit and the report shall be submitted no later than the 7<sup>th</sup> of each month within the Contractor's monthly OH&S Report.

### 24.1.23 Special Instructions:

24.1.23.1. For brand new machine ASLI can be permissible for first 6 months after mobilization as per OEM

24.1.23.2. ALSI Calibration to be done if there is any change/modification/break down/incident etc. to the crane and other applicable P&M.

24.1.23.3. Package to package mobility of heavy plant & machinery is not advisable in CMRL projects, if such resources are to be shared between two different packages, the receiving package should possess all the relevant documents such as third-party certificates mentioning name of the package and ASLI calibration certificate.

24.1.23.4. Heavy Machinery Operators Qualification criteria:

a). Contractor shall engage qualified operator of cranes of all types by following considerations:

❖ Completion of 21 years of Age

❖ Five years of relevant experience with valid license.

b). Contractor shall engage qualified piling rig/trench cutter/hydraulic grab operators by following considerations:

❖ Completion of 21 years of Age

❖ Five years of experience with valid license.

c). Contractor shall engage qualified operators with valid LMV/HMV licenses for Backhoe loader, Excavator and earth moving equipment's, etc.. by following considerations:

❖ Completion of 21 years of Age

❖ Three years of experience with valid license.

d). Criteria for selection of type of license consideration for operator, based on the unladen weight of equipment of vehicle.

i. Above 7.5 Ton-HMV License to be considered for operator.

ii. Below 7.5 Ton-LMV License to be considered for operator.

24.1.23.5. Contractor should provide Competency training on Safe operation of machinery and defensive driving for all the operators involved in plant & machinery/construction machinery

such as crane, rig, trench cutter, grab, Excavators, backhoe loaders, man lifters, transit mixer, muck dumper, trailer, etc. prior to engaging for work activities at CMRL-Project sites.

24.1.23.6. Third Party Agency evaluation of training records to be maintained with photo and video evidence which will be subjected to verification by Employer/Engineer Team.

## **25. Tunnelling Operations**

### **25.1 Detailed planning for Occupational Health and safety**

#### **25.1.1 General**

25.1.1.1 The aim of this manual is to encourage all parties to work together to reduce Occupational Health and safety risk. At the detailed planning stage, the project risk assessment will highlight areas of risk. These should be addressed by the parties to the process. Throughout the detailed planning process, the risk assessment should be revised to enable the parties to develop a construction-stage health and safety plan tailored to the specific circumstances of the project.

### **25.2 Parties to the processes**

25.2.1 The following parties to a tunnelling project have duties and responsibilities.

- a) the principal contractor responsible for the execution of the construction stage, the coordination of occupational health and safety activities on site and the effective management of risks during all construction work;
- b) The Contractors who have responsibility for the safe execution of part or parts of the works.
- c) the designer with responsibility for the design and specification of the works, whether temporary or permanent;

25.2.2 Other parties, depending on the method of procurement, can include:

- 1) plant and equipment manufacturers and suppliers. These bodies have duties under other legislation to supply plant and equipment that is safe;
- 2) the engineer who may, under the contract, have responsibility for the design and supervision of the works and for checking compliance with the terms of the contract.

### **25.3 Integral nature of design and construction**

25.3.1 The interdependence of the design and construction and their appropriateness to the ground and environmental conditions is more safety-critical in the execution of underground work than in any other construction activity. It follows that there should be authoritative engineering continuity through the stages of planning, investigation, design and construction. This is more effectively achieved by the involvement of a single organization throughout. If, however, there is a change in the engineering direction of the works, plans shall be devised to ensure that the essential continuity is maintained and that the total planning, design and construction process is not fragmented.

### **25.4 Pre-construction-stage organization**

25.4.1 As soon as practicable during the planning process, a Chief Safety Manager shall appoint by the contractor. The Safety Manager shall ensure that foreseeable risks likely to arise from shaft sinking or tunnelling work, and which might affect health and safety.

25.4.2 No contractor is allowed to carry out any tunnelling work unless they are able to demonstrate competence in the type of tunnelling work that is proposed and shows a willingness and ability to allocate sufficient resources to identify and manage risks effectively and to comply fully with the occupational health and safety legislation and the CMRL Contract requirements.

## **25.5 Construction stage**

### **25.5.1 Organization**

- a. The construction-stage health and safety plan is the fundamental document that ensures safe working throughout the construction of a tunnel. This plan shall be based on the information supplied by the client on the residual risks and should be prepared by persons suitably experienced in the tunnelling operations likely to be involved, including the types of plant and equipment to be used.
- b. The Engineer is obliged to ensure that the works do not commence until a satisfactory construction-stage health and safety plan has been produced by the principal contractor. At the same time, contractors should carry out an assessment of the risks posed to their employees, and should use this assessment in devising and implementing safe systems of work. The health and safety plan should define a clear chain of responsibility and channels of communication on engineering safety matters. This is particularly important in, for example, the choice of ground-support systems and decisions concerning safety at the tunnel face.
- c. The principal contractor shall be responsible under the Conditions of Contract for coordinating the activities of all other contractors and includes a requirement for the provision, accessibility and maintenance of welfare facilities. This needs particular consideration in long tunnels, deep shafts and in other remote locations. Where tunnel size permits, welfare facilities should be located underground, including on TBMs.
- d. A system for accounting for site personnel in an emergency shall be implemented and rigorously enforced. The system must accurately indicate the number and location of persons underground at all times. This information is essential to the emergency services in the event of an underground fire or other emergency. If the works have a single entrance/exit, a simple tally board can be adequate. For works with multiple entrances/exits, a more sophisticated system is necessary. Whichever system is used, it must be supervised and monitored, and all persons using the system must be instructed in its correct use.
- e. If the tunnelling operations form part of a multiple-contractor project, then the tally system used must be uniform across the entire extent of the underground works, and shall be built into the contract and safety case. The system must allow persons to be accounted for across the entire project site, irrespective of the entrance or exit by which they enter or leave.

- f. The contractor shall ensure
- i) Every compressed air system in a tunnel is provided with emergency power supply for maintained continued supply of compressed air as per Rule 155 of TBOCWR 2006.
  - ii) Watertight bulkhead doors are installed at the entrance of a tunnel to prevent flooding.
  - iii) Reliable and effective means of communication such as telephone or walkie-talkie are provided and maintained for arranging better effective communication at an excavation or tunnelling work as per Rule 136 of TBOCWR 2006.
  - iv) All portable electrical hand tools and inspection lamps used in underground and confined space at an excavation or tunnelling work is operated at a voltage not exceeding 24V.
  - v) Only flame proof equipment of appropriate type as per IS: 5571:2000 and or other relevant national /International standard is used inside the tunnel.
  - vi) Petrol or LPG of any other flammable substances are not used or stored inside the tunnel. Oxy-acetylene gas is not permitted for use in a compressed air environment.
  - vii) Temperature in any working chamber in an excavation or tunnelling work where workers employed does not exceed 29°C as per Rule 165 of TBOCWR 2006.

## **25.6 Statutory & Contractual controls**

25.6.1 OHS&E duties and responsibilities with respect to “construction work” are set out in the CMRL OHS&E Volumes 1-4 including the “relevant statutory provisions”, which are the various sets of Regulations and Rules that may apply to individual work activities. Administration is by the Engineer appointed by CMRL and enforcement is undertaken by the Inspectors of the Employer /Engineer.

25.6.2 All accidents and incidents, whether or not they result in personal injury or losses, must be reported to the Employer /Engineer. Careful records shall be kept by the contractor, and these records, together with the results of any associated investigation work and details of similar incidents from elsewhere, should be studied and analysed, and appropriate action taken.

## **25.7 Supervision and inspection**

25.7.1 All tunnelling operations shall be subject to appropriate supervision. Understanding, making engineering judgements, and taking action in response to varying ground conditions can be as vital to safe working conditions as the anticipation of hazards and risks. The contractor shall appoint appropriately qualified and experienced persons to monitor the site activities on their behalf, and to assist in overall promotion of OH&S. These appointments shall comply with the provision as directed in OHS&E Volume 1.

25.7.2 A team comprising of contractor’s Tunnel Manager and Senior Safety Manager(Tunnel) shall conduct a monthly Tunnel Audit and the report shall be submitted no later than the 7<sup>th</sup> of each month within the Contractor’s monthly OH&S Report.

## **25.8 Temporary works**

- 25.8.1 Temporary works of a substantial character are invariably necessary in the construction of shafts and tunnels. They can constitute an essential element of the permanent construction, as in the case of, for example, a pilot tunnel, a working shaft or the framing to form an opening. They can also be structurally independent incidental works, such as working gantries, access systems, thrust pits and walls.
- 25.8.2 It is essential that OH&S issues relating to these works be taken into account early in the design stage. Appropriate drawings, calculations, specifications and any contributions to the construction-stage OH&S plan shall be prepared and submitted to the Employer /Engineer for notice of no objection. Where any failure of these temporary works would have a significant adverse effect on the OH&S of persons or on the environment, they shall be the subject of an independent design check by the contractor. Once checked, changes should only be made after direct consultation with the appropriate designer.
- 25.8.3 The principal contractor shall appoint a person to ensure safety in the design, checking, construction and use of temporary works.

## **25.9 Records and reports**

- 25.9.1 The keeping of proper and adequate records of the physical aspects of the work is of the utmost importance. This is not only for safety during construction, but for safety in any necessary future maintenance and repair work. Such records shall be kept as part of the health and safety file/Operations and Maintenance Manual for the project. It must contain a longitudinal geological section record based on day-to-day records of the strata encountered, correlated with any borehole records or other records, and with any abnormal ground levels, tidal or river levels and rainfall. In addition, settlements and movements of the tunnel lining and supports, and the ground above and adjacent to the tunnel, shall be recorded. The records shall be made available for inspection at the request of the Employer /Engineer and formally submitted as instructed by the Employer /Engineer.

## **25.10 Recruitment of project personnel**

- 25.10.1 All persons employed should have a high standard of physical fitness. All persons working underground and all plant operators and banksmen be not less than 21 years old. Applicants for engineering and management posts must be assessed on their knowledge of and expertise in OH&S issues relevant to the project. Key Personnel shall be subject to notice of no objection by the Employer /Engineer.

## **25.11 Competence and training**

### **25.11.1 Competence**

- 25.11.2 All persons underground must be competent for the environment in which they are working and for the work tasks and activities they are required to carry out. Engineers, managers and supervisors should be competent both with respect to the work under construction, and in the techniques of management, communications and supervision. Evidence of competence such as the achievement of recognized qualifications should be sought. This could include, for example, qualifications in project management or

health and safety management; or further education qualifications in tunnelling-related subjects.

## **25.12 Training**

25.12.1 It is a legal requirement for all persons at work to be given appropriate training in health and safety related to the risks they might encounter at work. Special training requirements apply, e.g. to first-aiders, plant operators, and, in the case of compressed-air tunnelling, to all those exposed to pressurized atmospheres. Induction training must be given before any person starts work underground, whether as a new employee or as a person new to the project. Persons who are new to the industry are particularly vulnerable, and their induction training and the degree of supervision of their work, particularly initially, shall reflect this. Induction training should cover:

- likely hazards and risks;
- the devised safe methods of working;
- site rules and prohibited activities;
- the use of protective equipment;
- communications and responsibilities for health and safety;
- action in the event of any emergency, particularly egress from underground workplaces.

25.12.2 Persons employed for the maintenance of mechanical plant should have served an appropriate apprenticeship or similar training; however, a reasonable proportion of these persons may be trainees. Persons employed in the installation, operation and maintenance of electrical circuits and equipment should be graded in accordance with the current electrical contracting industry, and should be suitably trained for the work. They must have access to professional electrical engineering advice and supervision.

25.12.3 Persons trained in the use of specialized tunnelling plant shall also be experienced in general procedures for tunnelling and must be familiar with both OH&S instructions and operating instructions provided by plant manufacturers and suppliers. NOTE: Specialized tunnelling plant includes items such as TBMs, segment erectors (including winches), concrete spraying equipment, grout pans, drilling and coring equipment and pipe jacking pressure systems.

25.12.4 Toolbox talks are a valuable method of on-the-job-briefing and for imparting and updating health and safety knowledge. Such talks shall be held regularly, and should endeavour to cover each major work activity. The talks shall be repeated whenever circumstances in the workplace change significantly or when new hazards are anticipated or become evident. Visitors to underground projects should be given appropriate advice and training.

## **25.13 Personal protective equipment**

### **25.13.1 General**

25.13.1.1 Reliance solely on personal protective equipment should be regarded as a last resort. However, in tunnel construction works, the wearing of some personal protective equipment is inevitable. The contractor shall carry out risk assessments for PPE.

### **25.3.2 Head protection**



25.3.2.1 All persons on site shall wear protective headgear conforming to BS EN 397:1995 or equivalent Indian Standard at all times.

### **25.13.3 Foot protection**

25.13.3.1 Strong boots, with steel sole plates and toe protection, conforming to BS EN ISO 20345-2004 or equivalent Indian Standard, shall be worn at all times. In wet tunnels, heavy rubber knee boots with sole plates and toe protection may be required. Boots without toe and sole protection shall not be considered acceptable by the Employer / Engineer.

### **25.13.4 Hand protection**

25.13.4.1 Gloves of appropriate material shall be worn where a person is at risk from cuts, abrasions, burns or contact with corrosive or irritant substances. Elbow-length gloves should be worn when grouting or handling corrosive or irritant substances. Barrier creams shall be made available and used by site personnel.

### **25.13.5 Eye protection**

25.13.5.1 Safety goggles, face shields and fixed shields shall be provided and used as appropriate for protection against flying particles, hazardous substances or injurious light rays. Eye protection must always be worn by site personnel engaged on sandblasting, shot-blasting, pressurized air and water cleaning, spraying concrete, grinding or chipping metal, cutting, burning and welding, grout mixing and placing. It shall also be worn where there is a foreseeable risk of injury while breaking, cutting or drilling concrete or stone or similar material, and by persons injecting liquids under pressure.

### **25.13.6 Whole body protection**

25.13.6.1 Protective clothing shall be provided and used where appropriate to the risk, e.g. in wet conditions, when burning or welding or when handling hazardous substances.

### **25.13.7 Respiratory protection**

25.13.7.1 Where there is a risk of respiratory damage, ventilation shall be employed to extract or dilute any hazardous materials. The use of respiratory protective equipment shall only be employed as a last resort and shall be demonstrated through the contractors risk assessment of the task and environment.

### **25.13.8 High-visibility clothing**

25.13.8.1 Suitable high-visibility clothing conforming to BS EN 471-2003 & A1-2007 or equivalent Indian Standard shall be worn by all persons on site.

### **25.13.9 Hearing protection**

25.13.9.1 Guidance on the selection and care of hearing protection is given in Section 19.

### **25.13.10 Self-rescuers**

25.13.10.1 All persons underground shall have immediate access to a self-rescuer as described in 14.10 and be trained in their use. Records of such training must be kept for inspection by the Employer/ Engineer.

### **25.13.11 First aid provision and procedures**

#### **25.13.11.1 General**



25.13.11.2 It is essential that persons trained in first aid and capable of responding rapidly to any incident be available on each shift during working hours. All persons shall be informed of the first aid arrangements of the project.

25.13.11.3 All contractors shall train their on-site supervisory personnel in managing emergencies involving first aid. All personnel should be told that, in the event of serious injury, a casualty shall be moved only by a trained first-aider, unless there is the immediate risk of further injury. A minimum of 1 first aider shall be available at all times per 50 persons or part thereof.

25.13.11.4 Good communications are essential between the working areas and the surface facilities. A clear plan of action shall be formulated to speed up the transfer of any injured persons from working areas and to ensure that ambulances can reach shaft tops or other access points quickly. Lifting arrangements in shafts should take this into account. Clear instructions should be given to all persons on the procedures to be adopted for evacuating tunnels in an emergency and for ensuring that injured persons are not left behind or unaccounted for.

#### **25.13.12 First aid facilities**

25.13.12.1 Sufficient first aid boxes should be provided in accordance BOCW Central Rules 1998. First aid boxes shall be made of suitable material, and designed to protect the contents as far as possible from damp and dirt. They should be clearly identified; the recommended marking is a white cross on a green background. Boxes should be readily accessible to working areas and should be in the charge of designated first-aiders on each shift. Appropriate eye wash facilities shall also be provided.

#### **25.13.13 First aid rooms**

25.13.13.1 A properly constructed and equipped first aid room shall be provided as per BOCW Central Rules 1998 to be used for treatment and rest. It should be in the charge of a person trained in first aid and should be available during all working hours.

#### **25.13.14 First aid training**

25.13.14.1 The CMRL OSHE Conditions of Contract require persons with first aid responsibilities to have successfully undergone a course of training, and to be in possession of a current certificate issued by an organization approved by CMRL.

#### **25.13.15 Stretchers**

25.13.15.1 Stretchers (complete with blankets) suitable for the confined space of a tunnel shall be provided and maintained. They must be readily accessible for use in working areas in an emergency, and should be protected against dirt and damp. In particular, where access to a tunnel is by a shaft, stretchers should, where practicable, be stored at tunnel level. Appropriate means of transporting an injured person to the surface must be provided.

#### **25.13.16 Ambulances**

25.13.16.1 Ambulance services BOCW Central Rules 1998 should be notified of the location of the site and the nature of the work to be carried out. All employees shall be made aware of the procedure for calling an ambulance during the site specific induction.

25.13.16.2 Notices shall be posted on site naming the first-aiders and giving instruction on the making of emergency calls. A notice giving the address of the hospital dealing with emergencies must be prominently displayed.

#### **25.13.17 Transport underground**

25.13.7.1 Consideration as determined within a risk assessment, shall be given to the provision of a rescue skip, train or dedicated vehicle to expedite the rescue of casualties from long tunnels.

### **25.14 Procedures**

25.14.1 The Contractor shall develop occupational Health and Safety procedures and methods of working to be adopted during the course of tunnelling operations. These procedures shall include but not be limited to;

- (a) Shafts and Tunnels Entry Procedure. (Including visitors.)
- (b) Blasting operations.
- (c) Atmosphere Monitoring. (Oxygen Levels, Explosive Gases, Carbon Monoxide, Hydrogen Sulphide, Oxides of Nitrogen, temperature, humidity, dust etc.)
- (d) Portal Gantry Crane Operating Procedures.
- (e) Emergency Preparedness Plan for the Shaft and Tunnels. (Including liaison with the Emergency Services.)
- (f) Work Train Operating Procedure.
- (g) Tunnel Boring Machine Cutter Head Chamber Entry procedure.

A detailed method statement must be produced by the Contractor, and approved by the Employer / Engineer before the commencement of any tunnelling operations.

### **25.15 Sanitation and Drinking Water**

25.15.1 Toilets shall be effectively and regularly cleaned and disinfectants provided.

25.15.2 At least 5 litres of clean drinking water shall be provided per person employed on the shift. The water shall be sited near the portal and also inside tunnels over 500 metres in length. The water shall be contained in a clean container with a tight fitting lid.

25.15.3 Washing and cleaning facilities shall be provided for all workers near the portal.

### **25.16 Lighting**

25.16.1 The Contractor shall provide adequate lighting at the face and at any other point where work is in progress. A minimum of 110 lux shall be provided at the face, walkways and similar work areas.

25.16.2 Emergency lighting shall be installed at the working faces and at 100m intervals along the tunnel to help escape workmen in case of accidents.

### **25.17 Ventilation**

25.17.1 The Contractor shall make provision for adequate ventilation of all shafts and tunnels. The ventilation shall be sufficient to ensure proper dispersal of any dust or fume.

### **25.18 Protection Against Fire**

25.18.1 As far as practicable, combustible materials shall not be used in the construction of any room or recess containing electrical apparatus.

25.18.2 No flammable material shall be stored in any part of the tunnel unless it is contained in suitable flameproof containers.

25.18.3 An adequate supply of suitable first aid firefighting equipment shall be kept at convenient locations throughout the tunnel. This equipment shall be tested at least once a month and records kept available for inspection by the Employer / Engineer.

## **25.17 Warning Signals**

25.18.1 The contractor shall install a suitable system of warning signals for the movement of plant and materials within shafts and tunnels.

25.18.2 The system shall be checked daily immediately prior to the commencement of tunnelling work under the supervision of a responsible person.

25.18.3 The Contractor shall make detailed emergency warning signals for cases of fire, tunnel collapse etc.

## **26. Blasting Operations**

### **26.1 Authorisation for Blasting**

26.1.1 The Contractor shall ensure that all blasting operations will only be permitted following consultations with the relevant authorities and subsequent issuing of the permission to blast permits. The Employer / Engineer must also give his consent in writing before any blasting operations take place.

26.1.2 All blasting shall be conducted under the direct supervision of a Licensed Shot firer.

### **26.2 Risk Assessment and Method Statements**

26.2.1 The Contractor shall produce a detailed hazard and risk assessment and an in depth method statement for amongst others the following elements:

- (a) Type of explosives to be used.
- (b) Anticipated effects of vibration on nearby structures.
- (c) Blasting patterns.
- (d) Delivery of the explosives.
- (e) Transportation and storage of explosives on site.
- (f) Drilling and charging of holes.
- (g) Warning sirens.
- (h) Measurement of Vibration
- (i) Provision of sentries.
- (j) Use of blast screens.
- (k) ALL CLEAR.
- (l) Ventilation following blasting.
- (m) Atmosphere monitoring.
- (n) Procedure for miss-fires.

## **27. Demolition**

### **27.1 General**

- 27.1.1 The Contractor shall ensure that all demolition works shall be carried out in a controlled manner under the management of experienced and competent supervision.
- 27.1.2 Prior to any demolition commencing, a survey shall be conducted to identify if there are any hazardous materials present, for example the presence of materials such as asbestos and lead.
- 27.1.3 If any hazardous materials are found, then consideration shall be given as to whether they shall need to be removed by a Specialist Agency or Sub-contractor prior to the main demolition works commencing.
- 27.1.4 Before the demolition commences all relevant notifications will need to be given to the local authorities and media.
- 27.1.5 Measures for protection to the public shall be required to be put into place in order to give protection from any possible falling debris and dust generation.
- 27.1.6 All power supplies and services shall be disconnected before any demolition work commences.

## **28.False work / Formwork**

### **28.1 General**

- 28.1.1 The contractor shall ensure that all false work / formwork has been properly designed and is suitable for the purpose.
- 28.1.2 All designed false work / formwork shall be erected in strict accordance to the design.
- 28.1.3 Prior to the loading and subsequent striking of false work / formwork, permission shall be obtained from the Contractor's Designer and Engineer who shall both inspect and sign off on the structure in person.
- 28.1.4 Adequate provision shall be made on the working platforms for the concrete placement operations, these shall include locations for vibrators and the unobstructed movement of personnel controlling the rubber hose during the concrete pumping operations or the concrete skip during any skipping operations.
- 28.1.5 The Contractor should use the following checklist to check that false work / formwork is being used safely;
  - (a) Have the design and the supports for shuttering and false work / formwork been checked?
  - (b) Is it being erected safely from steps or proper platforms?
  - (c) Are the props plumb and properly set out?
  - (d) Are the bases and ground conditions adequate for the loads?
  - (e) Are the correct pins used in the props?

- (f) Are the timbers in good condition?
- (g) Is it inspected by a competent person against the agreed design before permission is given to pour concrete?

## **29.Piling and Diaphragm walls**

### **29.1 General**

- 29.1.1 The contractor shall prepare safe systems of work and method statements for all work concerned with piling and diaphragm walls. He shall take the following points into consideration.
- 29.1.2 Piling Rig is not a lifting equipment shall not be used for any lifting operation.
- 29.1.3 Any excavated piles or panels shall not be left unattended, unless they are adequately fenced around to prevent accidental entry into the immediate vicinity of the pile or panel.
- 29.1.4 Because of the use of heavy plant and equipment in generally congested work areas then trained banksmen shall be deployed to control the movement of the plant and personnel interface.
- 29.1.5 All lifting operations shall be conducted in accordance with the requirements as detailed in Section 18 Lifting Operations.
- 29.1.6 Calcium Oxide shall not be used for stabilizing the excavated spoil as it is an acute irritant, unless an agreed method statement has been produced.
- 29.1.7 A method statement shall be produced by the Contractor, which details the process for grab retrieval in the event of a grab becoming detached during the course of a pile or panel excavation.
- 29.1.8 A method statement shall be produced by the Contractor, which details the process for stop end recovery.
- 29.1.9 Wheel washing facilities shall be available on the sites for washing down the spoil removal trucks and the concrete delivery vehicles.
- 29.1.10 Bentonite and polymer storage tanks shall be bunded around to retain any unintentional and uncontrolled spillage.
- 29.1.11 The contractor shall submit to the Employer / Engineer, for approval, proposals for the treatment of Bentonite slurry and its subsequent disposal.
- 29.1.12 No Bentonite spillage shall be allowed on any roads.
- 29.1.13 Regular site cleaning shall be carried out at all work-sites.
- 29.1.14 The Contractor as part of his Emergency Plans shall develop procedures for the collapse of piles and diaphragm walls.
- 29.1.15 Install automatic water sprinkler system to wet vehicular access.
- 29.1.16 Piling rig should be placed upon the 20mm steel plate for the piling operation

## **30. Work Adjacent to Live Railways**

### **30.1 General**

30.1.1 Whenever work is to be conducted in close proximity to the live railways then the following measures shall need to be addressed:

- (a) The laws of India, State rules shall be followed.
- (b) No persons are allowed to encroach onto the railway unless specific authority has been given by the owner.
- (c) Adequate protection shall be followed. (Provision of Block Inspectors, Flagmen and Lookouts.)
- (d) All persons shall wear high visibility clothing at all times.
- (e) Any induction training requirements of the railway owner shall be strictly observed

## **31. Work Adjacent to Live Roadways**

### **31.1 General**

31.1.1 Whenever working adjacent to any live roadways then the following aspects shall be considered.

- (a) Close liaison with the Police and Municipal Authorities.
- (b) Production of an agreed traffic management scheme in accordance with the local traffic laws. (Barriers, signs, lights and road markings.) this shall include adequate provision for pedestrians.
- (c) Provide traffic marshal /Traffic Controller high reflective safety vest fitted with led lights by all personnel engaged in the activities.
- (d) Traffic Marshals shall be appointed and deployed to ensure that all road movement is carried out safely.
- (e) Ensures that traffic Marshal/Traffic Controller is available at all times of the day /Night to alert motorists of the temporary civil works ahead.
- (f) Age of the traffic Marshal/Traffic Controller shall not be more than 55 years old at time of appointment at site.
- g). Recommended minimum age criteria for traffic marshal is 21 years old and he should have competency training certificate from employer/engineer approved third party agency before deployment at site.

## **32. Personal Protective Equipment**

### **32.1 General**

32.1.1 The Contractor shall at all times keep and maintain an adequate supply of suitable personnel protective equipment which shall be readily available for use at all times on the sites, and would include amongst others the following items:

- (a) Safety Helmets.
  - (b) Hearing Protection.
  - (c) Respiratory Protection.
  - (d) Eye Protection.
  - (e) Protective Gloves.
  - (f) Safety Footwear.
  - (g) High Visibility Clothing to BS EN 471 Class 3 standard
- 32.1.2 PPE's shall be kept available for use by all staff, workers and authorised visitors to the sites.
- 32.1.3 The Contractor shall remove from the site any worker who consistently refuses to wear the appropriate personal protective equipment.
- 32.1.4 The Contractor shall implement a hand protection programme subject to the acceptance of the Employer /Engineer. The programme will:
- a) Identify activities on site that can cause hand injuries;
  - b) Propose safety interventions such as engineering or administrative measures to reduce the hazard;
  - c) Select, provide and maintain suitable hand protection devices and supervise their use;
  - d) Review and monitor the programme to test its effectiveness.
- 32.1.5 where hand gloves are used as added protection against hand injury, Contractor shall assess the work hazards and ensure suitable hand gloves are provided and used. Latex coated gloves shall be used for general works only.
- 32.1.6 The Contractor shall provide, maintain and enforce the usage of PPE for all the personnel on site at all times. The following PPE shall be compulsory on site:
- a) Safety helmets with chin strap conforming to BS EN or IS standards.
  - b) Safety footwear with steel toe cap conforming to BS EN or IS standards
  - c) High-visibility vest/clothing conforming to BS EN or IS standards.
  - d) Safety belts shall be provided for restraining falls or safety harness for fall protection. Safety belts and harness conforming to BS EN or IS standards
  - e) Respirators / dust masks of the appropriate standard shall be provided for activities generating dust or fume.
- 32.1.7 The Contractor shall maintain and update all PPE issuance records.

### **33. Medical Facilities**

33.1.1 Physical fitness of workmen

33.1.2 The contractor shall ensure that his employees/workmen subject themselves to such medical examination as required under the law or under the contract provision and the record of the medical examination of every building worker employed by the contractor shall



maintained in a register in Form XII under Schedule-XIII of TBOCWR 2006 and such register shall be made available to the Employer/Engineer, on demand

33.1.3 The contractor shall not permit any employee/workmen to enter the work area under the influence of alcohol or any drugs.

### **33.2 Medical Examination**

33.2.1 The contractor shall arrange a medical examination of all his employees including his sub-contractor employees employed as drivers, operators of lifting appliances and transport equipment before employing, after illness or injury, if it appears that the illness or injury might have affected his fitness and, thereafter, once in every two years up to the age of 40 and once in a year, thereafter.

- i) The Contractor shall maintain the confidential records of medical examination or the physician authorised by the Employer/Engineer.
  - ii) No building worker is charged for the medical examination referred rule No 81 and rule No 223 of TBOCWR 2006 and the cost of such examination is borne by the contractor, employing such building worker;
  - iii) The medical examination shall include Full medical and occupational history. Certificate of medical examination shall be maintained by the contractor as per Form-XI under Schedule-XIII of TBOCWR 2006.
- a) Clinical examination with particular reference to
- i) General Physique;
  - ii) Vision: - Total visual performance using standard orthorator like Titmus Vision Tester should be estimated and suitability for placement ascertained in accordance with the prescribed job standards.
  - iii) Hearing: - Persons with normal must be able to hear a forced whisper at twenty-four feet. Persons using hearing aids must be able to hear a warning shout under noisy working conditions.
  - iv) Breathing: - Peak flow rate using standard peak flow meter and the average peak flow rate determined out of these readings of the test performed. The results recorded at pre- placement medical examination could be used as a standard for the same individual at the same altitude for reference during subsequent examination.
  - v) Upper Limbs: - Adequate arm function and grip
  - vi) Spine: - Adequately flexible for the job concerned.
  - vii) Lower Limbs: - Adequate leg and foot flexibility.
  - viii) General: - Mental alertness and stability with good eye, hand and foot coordination.
  - ix) Any other tests which the examining doctor considers necessary.

33.2.2 If the contractor fails to initiate and continue medical examinations as stated previously, the Employer / Engineer has the right to appoint a medical examiner through an agency and deduct the cost and overhead charges against the contractor.

### **33.3 Occupational Health Centre**

33.3.1 The contractor shall ensure the provision of a site occupational health centre. This may be mobile or static however, both must be maintained in good order and complete with facilities as per the Schedule IX, Schedule-X of TBOCWR 2006.

### **33.4 First Aid Bases**

33.4.1 The Contractor shall establish a First Aid Base, in accordance with the Employer Requirements, at each of his principal work areas. If during the life of the contract the Contractor's principal work area moves from one location to another, the Contractor shall be required to move his First Aid Base.

33.4.2 If the Contractor operates more than one principal work area, he will be required to have a First Aid Base at each of his principal work areas.

33.4.3 The First Aid Base shall consist of as a minimum;

- (a) A treatment room fitted with two treatment couches,
- (b) A hand wash basin with running water;
- (c) Lockable cupboards to contain sufficient medical supplies;
- (d) Bed.
- (e) Six Chairs with footrests
- (f) Desk and chair.
- (g) Six Stretchers (Which can be lifted and lowered by a crane.)
- (h) Pillows and blankets.
- (i) Refuse containers.
- (j) Medical dressings. (Bandages, plasters, antiseptic wipes.)
- (k) Eye irrigation sterile solution.
- (l) Paper towels.
- (m) Disposable gloves.

33.4.4 The first-aid unit shall be provided with air conditioning and shall be kept in a clean and tidy state at all times.

### **33.5 Medical Staff**

33.5.1 A qualified Doctor, Nurse and assistant Nurse shall be in attendance at the first aid base during all times when work is being undertaken on the site.

### **33.6 Ambulance**

33.6.1 A fully equipped ambulance and driver shall be provided at the first aid base during all working hours. The ambulance shall be equipped with the articles specified in Schedule-IV of TBOCWR 2006.

### **33.7 First Aid Boxes**

33.7.1 Portable first aid boxes will be maintained and is equipped with the articles specified in Schedule-III of TBOCWR 2006 at each local site offices and work locations where 20 or more persons work at a time. Every first-aid box or cupboard is distinctly marked 'FIRST AID'.

33.7.2 In each site office and location one employee, suitably trained in first aid, should be available at all working hours for the purpose of attending to emergencies.

### **33.8 HIV/ AIDS prevention and control**

33.8.1 The contractor shall also extend necessary organizational support to the appointed agency for the effective implementation HIV/AIDS programme for workmen of the Contractors.

### **33.9 Prevention of mosquito breeding**

33.9.1 Measures shall be taken to prevent breeding at site. The measures to be taken shall include:

- i) Empty cans, oil drums, packing and other receptacles, which may retain water shall be deposited at a central collection point and shall be removed from the site regularly.
- ii) Still waters shall be treated at least once every week in order to prevent mosquito breeding.
- iii) Contractor's equipment and other items on the site, which may retain water, shall be stored, covered or treated in such a manner that water could not be retained.
- iv) Water storage tanks shall be provided.

33.9.2 Posters in Hindi, Tamil and English, which draw attention to the dangers of permitting mosquito breeding, shall be displayed prominently on the site.

33.9.3 The contractor at periodic intervals shall arrange to prevent mosquito breeding by fumigation / spraying of insecticides. Most effective insecticides shall include SOLFAC WP 10 or Baytex, The Ideal Larvicide etc.

### **33.10 Alcohol and drugs**

33.10.1 The contractor shall ensure at all times that no employee is working under the influence of alcohol / drugs which are punishable under Govt. regulations.

33.10.2 Smoking at public worksites by any employee is also prohibited unless suitable amenities are provided in accordance with Govt. regulations.

33.10.3 Alcohol testing shall be carried out for all workers and the report should be submitted on monthly to the Employer / Engineer.

## **34. Fire Precautions**

### **34.1 General**

34.1.1 The Contractor shall be responsible for supplying and maintaining adequate fire precaution facilities on all his sites. The following minimum standards should be adhered to.

34.1.2 The Contractor shall ensure that specially trained personnel are available to deal with

fires due to electrical causes, gas explosions etc.

- 34.1.3 A good standard of housekeeping shall be maintained at all times on the sites.
- 34.1.4 No accumulations of rubbish shall be allowed to gather.
- 34.1.5 Combustible scrap and other construction debris should be disposed off site on a regular basis. If scrap is to be burnt on site, the burning site should be specified and located at a distance no less than 12 metres from any construction work or any other combustible material.
- 34.1.6 Signage shall be erected at prominent positions showing the correct use of portable first aid fire extinguishers.
- 34.1.7 Emergency Plan and Fire plan shall be prepared and issued. Mock drills should be held on a regular basis to ensure the effectiveness of the arrangements.
- 34.1.8 The Contractor shall ensure that all procedures, precautionary measures and safety standards stipulated in the Fire Plan are implemented, communicated and complied with by all workers including sub-contractors and Interfacing Contractors.
- 34.1.9 The Contractor shall review and ensure the adequacy of the Fire Plan as the Works progress.
- 34.1.10 The Contractor shall carry out monthly checks of fire fighting equipment and test all alarms and detection devices installed on site. Tags/stickers shall be provided to indicate the monthly checks.
- 34.1.11 The Contractor shall conduct weekly inspections of escape routes, fire brigade access, fire fighting facilities and work areas to ensure that the requirements stipulated in the Fire Plan are complied with.

## **34.2 Fire Fighting Equipment**

- 34.2.1 At various locations around the site clearly visible fire points shall be established for use in an emergency and each fire point should have available as a minimum the following type of equipment:
  - (a) Dry Powder Extinguisher.
  - (b) Water Type Extinguisher.
  - (c) Bucket of Sand.
- 34.2.2 Recharging of fire extinguishers and their proper maintenance should be ensured and as a minimum should meet Indian National Standards
- 34.2.3 Water supply for firefighting purposes should be provided at the construction site. This may be in the form of static water tank of adequate capacity or a hydrant line with adequate water pressure at outlet points.
- 34.2.4 Sufficient number of fire hoses with branch pipes should be provided at site so that the fire can be controlled until the arrival of the Fire Brigade.
- 34.2.5 The contractor shall need to give consideration to the provision of adequate firefighting arrangements within the underground and tunnelling operations including the provision

of Fire Service compatible hose connections and emergency lighting

34.2.6 The Telephone Number of the local fire brigade should be prominently displayed near each telephone on site.

34.2.7 Supervisors and workmen at the site should be trained in the use of firefighting equipment provided at the site.

### **34.3 Storage of Flammable Liquids**

34.3.1 All flammable liquids shall be kept in a secure fire resistant store protected from electrical sparks welding sparks open flames and smoking.

34.3.2 Only such amounts of flammable liquids should be issued as are required for immediate use. Cans for carrying flammable liquids should be leakproof and properly stoppered and clearly marked “FLAMMABLE LIQUID”.

34.3.3 Rags soaked in paints, kerosene and other flammable liquids should be disposed of daily under supervision. Large quantities of such rags should not be allowed to accumulate.

34.3.4 All Diesel fuel storage tanks shall be bunded around in order to control any spillage or leakage that may occur.

34.3.5 All drivers of vehicles, foreman, supervisors and managers shall be trained on operating the fire extinguishers and firefighting equipment

34.3.6 All lifting appliances’ driver cabin should be provided with a suitable 6kg portable fire extinguisher.

34.3.7 If the Contractor intends to store petroleum more than the legal requirements and flammable materials on site, he shall obtain a storage licence from the state government and a copy of the licence shall be submitted to the Employer /Engineer.

34.3.8 The Contractor shall not store petrol on site. If they want to store they should obtain licence from the govt of Tamil Nadu

34.3.9 All diesel stored on site shall be kept in drums or in bulk tanks which in either case shall be located at a designated place away from any sources of ignition or open drain which does not lead to an interceptor, and shall be properly labelled. A “No Smoking” sign shall be displayed at the storage location and a charged fire extinguisher of correct type kept on standby.

34.3.10 All bulk diesel tanks shall be properly supported in an elevated position to facilitate gravity discharge. They shall stand within a bund constructed to contain a volume of 110% of the volume of the tank. There shall be no breaches in the bund wall, no material shall be stored within the bund and rain water collecting in the bund shall be regularly removed to prevent build-up. The inner face of the bund wall shall be coated with a chemical resistant material. A chemical resistant valve, which shall be closed at all times, except for releasing rainwater into a stormwater drain via an oil intercepting system, shall be installed at the outlet situated outside the bund, in accordance with the legal requirements.

- 34.3.11 All drums of diesel on Site shall be in good condition and shall be kept closed with a lid/cap when not being used. They shall be stored on end with the lid/cap uppermost and kept within a tray of sufficient volume to contain the contents of the largest drum in the case of accidental rupture, taking into account the presence of other drums within the tray.
- 34.3.12 Drums of diesel shall not be rolled along the ground. They shall be transported vertically chained on a trolley; or by a forklift fitted with a drum handling device and not standing unsupported on the forks or on a pallet; or by crane using a safe slinging technique.
- 34.3.13 Diesel shall be transferred from the storage drum to another container, or to the tank of plant/machinery using a hand pump wherever practicable and at all times a drip pan must be provided. Where the diesel container is light enough to be lifted by one person it can be poured out by hand, using a funnel to guide the liquid.
- 34.3.14 Any spillages of diesel shall straight away be absorbed using sand or other absorbent materials, which shall be disposed of as contaminated waste. On no occasion should diesel be allowed to enter the Site drainage system unless this is connected to an interceptor prior to the Site waste being discharged into the public sewer system.

## **35. Construction Machinery**

- 35.1 Construction machineries may include dumpers and dump trucks, lift trucks and telescopic handlers piling rigs, vibro hammers, rail welding equipment's, mobile elevating work platforms, cranes, tipper lorries, lorry loaders, skip wagons, 360° excavators, 180° backhoe loaders, crawler tractors, scrapers, graders, loading shovels, trenchers, side booms, pavers, planers, chippers, road rollers, locomotives, tankers and bowsers, trailers, hydraulic and mechanical breakers etc.

### **35.2 Safe worthiness certificate**

- 35.2.1 All construction equipment shall be in sound mechanical working condition, tested and certified every six months by either competent person under Factories Act or manufacturers' warranty in case of brand new equipment or authorized persons/ Chartered Engineer (Mechanical) / firms approved by the Employer / Engineer before induction to any site.
- 35.2.2 Every such certificate shall have the date of purchase, main overhauling undertaken in the past, any accident to the equipment, visual examination details, critical components safety check, list of safety devices and its working condition, manufacturer's maintenance checklist, past projects wherein the equipment's were used etc. as its minimum content.

### **35.3 Reverse Horns**

- 34.3.1 All Vehicles shall be fitted with audible reverse alarms and rear-view camera, should be maintained in good working condition. Reversing shall be done only when there is adequate rear-view visibility or under the direction of a trained banksman.



34.3.2 The rear-view camera display shall not be less than 170x100x10 mm in size and display image should be in good quality and visibility to operator.

#### 35.4 General operating procedures

- i) Drivers entering site shall be instructed to follow the safe system of work adopted on site. These shall be verbal instructions or, preferably, written instructions showing the relevant site rules, the site layout, delivery areas, speed limits, etc.
- ii) No passengers shall be carried, unless specific seating has been provided in accordance with the manufacturer's recommendations.
- iii) Working on gradients beyond the equipment's capability shall not be permitted.
- iv) Prevention of dumper and dump truck accidents should be managed by providing wheel stops at a sufficient distance from the edges of excavations, spoil heaps, pits, etc.
- v) The manufacturer's recommended bucket size must not be exceeded in excavators.
- vi) If excavators operating on a gradient which cannot be avoided, it must be ensured that the working cycle is slowed down, that the bucket is not extended too far in the downhill direction, and that travel is undertaken with extreme caution. A large excavator must never be permitted to travel in a confined area, or around people without a banksman to guide the driver, who should have the excavator attachment close in to the machine, with the bucket just clear of the ground. On wheeled excavators, it is essential that the tyres are in good condition and correctly inflated. If stabilizing devices are fitted, they should be employed when the machine is excavating.
- vii) When the front shovel of the 180° backhoe loaders is being employed, the backhoe attachment shall be in its "travel" position, with the safety locking device in place.
  - viii) When operating the backhoe in poor ground conditions, the stabilizers tend to sink into the surface of the ground, reducing stability. Therefore frequent checks shall be made for the stability of the machine. The loading shovel should always be lowered to the ground to stabilize the machine when the backhoe is employed.
- ix) The netting operation of the skip wagons should be carried out prior to lifting the skip to reduce the risks of working on the rear platform.
- x) If a tractor dozer is employed on clearing scrub or felling trees, it shall be provided with adequate driver protection from rollover and falling objects.
- xi) When two or more scrapers are working on the same job, a minimum distance of at least 25m shall be kept between them.

35.5 All wood working machines shall be fitted with suitable guards and devices such as top guard, riving knife, push-stick, guards for drive belts and chains, and emergency stop switch easily accessible by the operator.



### 35.6 Penalty

35.6.1 If any of the above clauses are not adhered, penalty shall be imposed as per relevant clause depending upon the gravity of the unsafe act and or condition.

## 36. Traffic Management

### 36.1 General

- 36.1.1 The Contractor shall provide, install and maintain all necessary traffic and directional signs, barriers, blinkers, rotating beacons, cones, lane markings etc. in accordance with the legal requirements. The Contractor must observe the minimum clearance required between the working area and the trafficked carriageway and ensure that all plants and materials do not intrude into any area reserved for pedestrians, cyclists or other traffic.
- 36.1.2 The Contractor shall regularly maintain the site road surfaces to keep them free of potholes, unevenness, etc. Mill and patch method shall be required to repair any uneven surface defects.
- 36.1.3 The basic objective of the following guidelines is to lay down procedures to be adopted by contractor to ensure the safe and efficient movement of traffic and also to ensure the safety of workmen at construction sites.
- 36.1.4 All construction workers must be provided with high visibility jackets with reflective tapes as most of viaduct / tunnelling and station works are either located above or under right-of-way. The conspicuity of workmen at all times shall be increased so as to protect from speeding vehicular traffic.
- 36.1.5 At each entrance to the site the Contractor shall erect a large billboard warning all persons who enter the site that they are required to wear the appropriate Personal Protective Clothing and that no unauthorised access is allowed.
- 36.1.6 Wherever the fence runs adjacent to the highway with no buffer-zones then the fence shall have traffic warning lights duly affixed to it.
- 36.1.7 Pedestrian route must be provided and separated to avoid mishap involving workers crossing the roads from different directions ,wherever the fence borders on pedestrian footpaths lighting shall be provided to illuminate the pedestrian routes. Additional precautions must be taken and flash lights / flag man required to be placed during foggy / adverse weather. The positioning of the fence-line shall not reduce the width of the pedestrian footpath to less than 900 mm in order to be able to accommodate disabled persons in wheelchairs.
- 36.1.8 Barricade Boards shall be washed at least once a week and repainted at least annually. All barricading boards shall be properly pegged (12 pegs for each board) & all inter connection between the adjacent boards (3 no's) shall tied.
- 36.1.9 The site fencing shall need to be inspected on a regular basis in order to ensure that the integrity of the fencing is maintained at all times as far as is practicable.
- 36.1.10 The guiding principles to be adopted for safety in construction zone are to;
- Warn the road user clearly and sufficiently in advance.

- ii. Provide safe and clearly marked lanes for guiding road users.
- iii. Provide safe and clearly marked buffer and work zones
- iv. Provide adequate measures that control driver behaviour through construction zones.
- v. Materials and equipment such as pile cages shall not be transported via Hydra on a public highway unless all access is closed to members of the public.
- vi. Impact protection vehicle(IPV) should be placed in advance while working on live road.
- vii. Vinyl(2x2.5meter) public /Traffic safety information poster shall be pasted on the each barricaded board.
- viii. Light mast should be placed at the start point of the barricaded board while working on live road.
- ix. Rope light shall be installed on all barricade board.
- x. Traffic digital sign boards shall be installed at the entry, exit, diversion, Warning, Road Crossing, Left/Right Carry Way Closed, Tunnel, Electric Cable, Speed Bump, Men at work, Pedestrian Crossing etc.
- xi. Pedestrian route shall be provided with top rail, midrail and the pipes shall be painted in reflective paints.

## **36.2 Legal permission**

- 36.2.1 In all cases, the contractor shall employ planned precautions specific to the location of the site works. Wherever operations undertaken are likely to interfere with public traffic, specific traffic management plans shall be drawn up and implemented by the contractor in consultation with the approval of local police authorities and/or the concerned metropolitan/civil authorities as the case may be.
- 36.2.2 Such traffic management plans shall include provision for traffic diversion and selection of alternative routes for transport of equipment. If necessary, the contractor shall carry out road widening before commencement of works to accommodate the extra load.
- 36.2.3 The primary traffic control devices used in work zones shall include signs, delineators, barricades, cones, pylons, pavement markings and barrier lighting.
- 36.2.4 The road construction and maintenance signs which fall into the same three major categories as do other traffic signs, that are Regulatory Signs, Warning Signs and Direction (or guidelines) Reflective Signs shall only be used. The IRC: 67 (Code of Practice for Road Signs) provide a list of traffic signs. The size, colours and placement of sign shall confirm to IRC: 67. The contractor shall submit Traffic Sign design to the Employer / Engineer for Notice of No objection.

## **36.3 Regulatory signs**

- 36.3.1 Regulatory signs impose legal restriction on all traffic. It is essential, therefore, that they are used only after consulting the local police and traffic authorities.

### **36.4 Warning signs**

- 36.4.1 Warning signs in the traffic control zone shall be utilised to warn the drivers of specific hazards that may be encountered.
- 36.4.2 The contractor shall place detour signage at strategic locations and install appropriate warning signs. In order to minimize disruption of access to residences and business, the contractor shall maintain at least one entrance to a property where multiple entrances exist.
- 36.4.3 A warning sign shall be installed on all secondary road which merges with the primary road where the construction work is in progress at a sufficient distance before it merges with the primary road so as to alert the road users regarding the 'Metro Work in Progress'.

### **36.5 Delineators**

- 36.5.1 The delineators are the elements of a total system of traffic control and have two distinct purposes:
- i. To delineate and guide the driver to and along a safe path
  - ii. As a taper to move traffic from one lane to another.
- 36.5.2 These channeling devices such as cones, traffic cylinders, tapes and drums shall be placed in or adjacent to the roadway to control the flow of traffic. These should normally be retro-reflectors complying with IRC: 79 - Recommended Practice for Road Delineators.

### **36.6 Traffic cones and cylinders**

- 36.6.1 Traffic cones of 500mm, 750mm and 1000mm high and 300mm to 500mm in diameter or in square shape at base and are often made of plastic or rubber and normally have retro-reflectorised red and white band shall be used wherever required.

### **36.7 Drums**

- 36.7.2 Drums about 800mm to 1000mm high and 300mm in diameter can be used either as channelising or warning devices. These are highly visible, give the appearance of being formidable objects and therefore command the respect of drivers. Drum delineators must be sufficiently illuminated.

### **36.8 Barricades**

- 36.8.1 Full height fence, barriers, barricades etc. shall be erected around the site in order to prevent the working area from the risk of accidents due to public vehicular movements. The barriers are also designed to protect road users from the dangers associated with general construction operation material handling and temporary structures.
- 36.8.2 The structure dimension of the barricade, material and composition, its colour scheme, CMRL logo and other details shall be in accordance with specifications laid down in tender document.
- 36.8.3 All barricades shall be erected as per the design requirements of the Employer/Engineer, numbered, painted and maintained in good condition and also Barricade in-charge maintains a barricade register in site. All barricades shall be washed at least once a week

and repainted at least annually. The contractor shall submit barricade design to the Employer / Engineer for Notice of No objection.

36.8.4 All barricades shall be conspicuously seen in the dark/night time by the road users so that no vehicle hits the barricade. Conspicuity shall be ensured by affixing retro reflective stripes of required size and shape at appropriate angle at the bottom and middle portion of the barricade at a minimum gap of 1000mm. Red light blinker and Reflective sticker plate should be placed at the top of each barricade.

36.8.5 The contractor shall ensure that all his construction vehicles plying on public roads (like dump trucks, trailers, etc.) have appropriate licenses to ply on public roads from the Govt. All drivers shall be medically fit and in possession of a valid and current driving licence as per the requirements of motor vehicles act as amended in 1994, the central motor vehicles rules, 1989 shall drive these vehicles on the public highway.

36.8.6 The contractor shall not undertake loading and unloading at carriageways obstructing the free flow of vehicular traffic and encroachment of existing roads by the contractor applying the excuse of work execution unless traffic management alternatives have been installed e.g. diversion.

36.8.6 The Contractor shall undertake Barricade Board maintenance work and painting etc. at the Engineer's request where concerns have been raised regarding the suitability of barricading board. Such Barricade Board maintenance shall be carried out with immediate effect.

### **36.9. Tow away vehicle**

36.9.1 The contractor shall make arrangements keeping tow away van / manpower to tow away any breakdown vehicle in the traffic flow without loosing any time at his cost.

### **36.10 Cleaning of roads**

36.10.1 The contractor shall ensure the cleanliness of roads and footpaths by deploying appropriate manpower to ensure efficiency in the cleaning task. Vehicle Control

36.10.2 Traffic Controllers shall be available for directing vehicles that are exiting the sites directly onto the live carriageways. Any vehicles entering the sites that are required to execute reversing manoeuvres shall do so under the strict control of a trained and designated banksman.

### **36.11 Spoil Removal**

36.11.1 Any vehicles leaving the sites carrying loads which are liable to produce airborne contaminants shall prior to leaving the site securely sheet the load over in order to effectively contain any dispersement during transportation on the public highway.

36.11.2 Vehicles exiting the site directly onto the live carriageway shall do so under the control of the clearly identified Traffic Controller.

36.11.3 Any vehicles that are required to reverse whilst on the site shall do so under the control of a trained banksman.

36.11.4 Any vehicles prior to leaving the site shall have their wheels washed and any loose material removed.

36.11.5 Any spoil that is removed from the work-sites shall be disposed of only at authorised dumping sites.

36.11.6 A team comprising of contractor's construction Manager and Traffic Engineer shall conduct a monthly Traffic Safety Audit and the report shall be submitted no later than the 7<sup>th</sup> of each month within the Contractor's monthly OH&S Report.

### **37. Housekeeping**

37.1 Housekeeping is the act of keeping the working environment cleared of all unnecessary waste, thereby providing a first-line of defence against accidents and injuries.

37.2 Contractor shall understand and accept that improper housekeeping is a primary hazard in any construction site and ensure that a high degree of house keeping is always maintained.

37.3 Housekeeping is the responsibility of all site personnel, and line management commitment shall be demonstrated by the continued efforts of supervising staff towards this activity.

37.4 General House keeping shall be carried out by the contractor and ensured at all times at the Work Site, Construction Depot, Batching Plant, Labour Camp, Stores, Offices and toilets/urinals. Towards this the Contractor shall constitute a special group of house keeping personnel. This group shall ensure daily cleaning at work sites and surrounding areas and maintain a register for inspection at the Employer /Engineer request.

37.5 Adequate time shall be assigned to ensure that good housekeeping is maintained. This shall be carried out by the housekeeping team.

37.6 The contractor shall be responsible to provide segregated containers for disposal of debris at required places and regular cleaning of the same.

37.7 Full height fence, barriers, barricades etc. shall be erected around the site in order to prevent the surrounding area from excavated soil, rubbish etc, which may cause inconvenience to and endanger the public. The barricade especially those exposed to public shall be aesthetically maintained by regular cleaning and painting as directed by the Employer / Engineer. These shall be maintained in one line and level.

37.8 All stairways, passageways and gangways shall be maintained without any blockages or obstructions. All emergency exits passageways, exits fire doors, break-glass alarm points, fire fighting equipment, first aid stations, and other emergency stations shall be kept clean, unobstructed and in good working order.

37.9 Lumber with protruding nails shall be either bent or removed and properly stacked. Un-packed wooden strips shall be safely stored and disposed.

37.10 All surplus earth and debris are removed/disposed off from the working areas to officially designated dumpsites. Trucks carrying sand, earth and any pulverized

materials etc. in order to avoid dust or odour impact shall be covered while moving. The tyres of the trucks leaving the site shall be cleaned with water, wherever the possibility of spillage on carriageways meant for regular road traffic exists.

- 37.11 No parking of trucks/trolleys, cranes and trailers etc. shall be allowed on roads, which may obstruct the traffic movement.
- 37.12 Roads shall be kept clear and materials like: pipes, steel, sand boulders, concrete, chips and brick etc.
- 37.13 Water logging or bentonite spillage on roads shall not be allowed. If bentonite spillage is observed on road endangering the safety of road users, the contractor shall be penalized as per Table -6 of Employer's Requirements OHS&E Volume 1.
- 37.14 Proper and safe stacking of material are of paramount importance at yards, stores and such locations where material would be unloaded for future use. The storage area shall be well laid out with easy access and material stored / stacked in an orderly and safe manner.
- 37.15 Flammable chemicals / compressed gas cylinders shall be safely stored.
- 37.16 Unused/surplus cables, steel items and steel scrap lying scattered at different places within the working areas shall be removed to predetermined locations(s).
- 37.17 All wooden scrap, empty wooden cable drums and other combustible packing materials, shall be removed from work place to identified location(s).
- 37.18 Empty cement bags and other packaging material shall be properly stacked and removed.
- 37.19 The Contractor shall ensure that all his sub-contractors maintain the site reasonably clean through provisions related to house keeping
- 37.20 The Contractor shall implement a 5S housekeeping method approved by the Employer /Engineer. The method shall be based on a Japanese quality management concept based on cyclical methodology. The 5S shall consist of Seiri (Sort/Organise), Seiton (Straighten/Orderliness), Seiso (Sweep/Cleanliness), Seiketsu (Standardise) and Shitsuke (Sustain/Discipline).
- 37.21 The Contractor shall transport waste material directly from the excavation to the approved dumping ground and shall discharge the waste directly into position. No stockpiles of waste are allowed on site.

## **38. Overhead Protection**

- 38.1 All contractors shall provide overhead protections as per TBOCWR 2006
- i) Overhead protection (Fan) should be erected along the periphery of every building which is under construction and the building height shall be 15m or above after construction.



- ii) Overhead protection shall be minimum 2m wide and the outer edge shall be 150mm higher than the inner edge and an angle not more than 20degree to its horizontal sloping into the building.
- iii) Overhead protection shall not be erected more than a height of 5m from the base of the building.
- iv) Areas of inadvertent hazard of falling of material shall be guarded or barricaded to prevent access and appropriately signed.

### **39.Slipping, Tripping, Cutting, Drowning and Falling Hazards**

- i) All places should be free from dust, debris or similar materials.
- ii) sharp projections or any protruding nails or similar objects shall be suitably guarded or shall even be avoided to make the place safe to work.
- iii) Contractor shall not allow workmen to work or use platforms, scaffolds/passageways or any walkways, which has water, or oil or similar substances spilt and has a slipping hazard, unless it is cleaned off or covered or sanded or saw dusted or make it safe with any suitable material.
- iv) When workers are exposed to areas where fall into water is possible, the contractor shall provide suitable and adequate equipment for saving the workers from drowning and rescuing from such hazard. If the Employer / Engineer considers, the contractor shall provide well-equipped boat or launch, manned with trained personnel at the work place.
- v) Open side or opening where worker, equipment, vehicle or lifting appliance may fall at a building or outside shall be guarded suitably except in places of free access by reasons of nature of work.
- vi) Suitable safety net shall be provided at places of material / man falling is possible in
- vii) accordance with national standards

### **40.Launching Operation**

- 40.1 The contractor shall take utmost precaution at all stages of a launching operation to include but not limited to; planning, establishing casing yard, casting segments & girders, transportation and load restrictions applicable, bridge and road conditions, fabrication and erection of launching girders if any, launching of segments / lifting of segments, pre-stressing, auto launching of girders and dismantling of launching girders.
- 40.2 The contractor shall prepare a comprehensive Method Statement for the launching operation, Particular reference shall be made to the provisions on working at height & heavy lift planning. As the process of launching/lifting may be required utilizing the public right-of-way, the safety of workers and the equipment serviceability is of paramount importance. The following general guidelines shall be adhered throughout the launching operation.
  - i) Necessary 'working platforms' and fall protection anchorage arrangement shall be



provided in the launching girder itself.

- ii) Provisions for mounting light fittings shall also be made available in the launching girder.
- iii) The workmen engaged in fabrication of reinforcement, concreting the segment shall be provided with necessary PPEs including compulsory hand protection gloves.
- iv) Casting and curing of segment / girder shall be undertaken under the direct supervision of the responsible engineer of the contractor.
- v) Trucks with valid registration, licence, safe worthiness certificate, Employer's approval certificate, and pollution under check certificate shall only be used for transport of segments
- vi) Drivers engaged for driving these trucks, shall receive refresher.
- vii) Drivers shall also have undergone proper medical examination as per relevant clause mentioned under 'Medical Facilities'.
- viii) The segments / girders shall rigidly secured to the truck with necessary wooden wedges and necessary red indicators/safety tapes provided so that the vehicle is clearly seen by other road users both in day / night time.
- ix) Every launching girder shall have a responsible engineer on duty all the time.
- x) All the time from erection to dismantling the area between the two piers wherein launching is in progress shall always be barricaded.
- xi) Unloading of segments / girders from trucks, lifting of loads, shifting of segments / girders, gluing shall be done under the direct supervision of the approved engineer of the contractor.
- xii) Auto launching shall be done only after approval from the Employer / Engineer. After every auto launching the stability of launching girder shall be ensured.
- xiii) The vertical deflection of launching girder shall be monitored at all critical stages like with/without loads and after every auto launching.
- xiv) A register containing all important operational details from erection to dismantling of launching girders shall be maintained and made available to the Employer / Engineer whenever called for.
- xv) Test certificate for all lifting gears including Macalloy bars shall be maintained at a location closer to the launching girder itself so that it can be referred during all inspections.
- xvi) Adequate lighting at all time shall be ensured in the entire area of operation.
- xvii) Access to drinking water & toilet shall be ensured to all workmen engaged for launching process.
- xviii) Proper access stairways shall be maintained for safe ascending and descending of workmen / engineers.

- 40.3 Non-adherence to any of the clauses mentioned above shall be viewed seriously by the Employer and penalty levied as per relevant clause.

#### **41. Machine and General Area Guarding**

- 41.1 The contractor shall ensure at the construction site all motors, cogwheels, chains and friction gearing, flywheels, shafting, dangerous and moving parts of machinery are securely fenced or legged. The fencing of dangerous part of machinery must not be removed while such machinery is in motion or in use.

#### **42. Manual Lifting and Carrying of Excessive Weight**

- 42.1 The contractor shall ensure that no worker lifts by hand or carries overhead or over his back or shoulders any material, article, tool or appliances of such a weight that may cause injury from either a single lift or repetitive lifting. A Risk Assessment regarding suitable loads to preventing injury must be carried out by the contractor.

| Person      | Maximum weight in kg. |
|-------------|-----------------------|
| Adult man   | 55                    |
| Adult woman | 30                    |

- 42.2 No building worker aided by other building worker shall lift or carry weight higher than or exceeding the sum of total of maximum limits set out for each building worker separately as mentioned in the table above.

#### **43. Hand Tools and Power Tools**

##### **43.1 General**

- 43.1.1 The contractor is wholly responsible for the safe condition of tools and equipment used by his employees and that of his sub-contractors.
- 43.1.2 Use of damaged hand tools shall be avoided and the contractor shall ensure all his hand tools used at his worksite are safe to work with or stored and shall also train his employees (including his sub-contractors) in their correct use.
- 43.1.3 All hand tools and power tools shall be duly inspected before use for signs of damage.
- 43.1.4 All hand tools and power tools shall have sufficient grip and the design specification on par with national/international standards on anthropometrics.

##### **43.2 Hand tools**

- 43.2.1 Hand tools shall include saws, chisels, axes and hatches, hammers, hand planes, screw drivers, crow bars, nail pullers, scaffolders podge etc.
- 43.2.2 The contractor shall ensure that,

- i) For crosscutting of hardwood, saws with larger teeth points (no. of points per inch) shall be preferred to avoid the saw jumping out of the work piece.
- ii) Mushroom headed chisels shall not be used in the worksite where the fragments of the head may cause injury.
- iii) Unless hatchet has a striking face, it shall not be used as a hammer.
- iv) Only knives of retractable blades shall be used in the worksite.
- v) No screwdrivers shall be used for scraping, chiselling or punching holes.
- vi) A pilot hole shall always be driven before driving a screw.
- vii) Appropriate PPE shall be issued to the workforce as dictated by the equipment and manufacturers recommendations.

### 43.3 Power tools

43.3.1 Power tools include drills, planes, routers, saws, jackhammers, grinders, sprayers, chipping hammers, air nozzles and drills.

43.3.2 The contractor shall ensure that

- i. Electric tools are properly grounded or / and double insulated.
- ii. GFCIs/ RCCBs shall be used with all portable electric tool operated especially outdoors or in wet condition.
- iii. Before making any adjustments or changing attachments, the tool shall be disconnected from the power source.
- iv. All drills shall have suitable attachments respective of the operations and powerful for ease of operation.
- v. Tools shall be specifically designed for that work that shall be undertaken.
- vi. Workers shall not stand on the top of the ladder to drill holes in walls / ceilings.
- vii. Electric plane shall not be operated with loose clothing.
- viii. Safety guards used on right angle head or vertical portable grinders must cover as per the wheel specifications shall be checked.
- ix. All power tools / hand tools shall have guards at their nip points.
- x. Leather aprons and gloves shall be used as an additional personal protection auxiliary to withstand kickback.
- xi. Push sticks shall be provided and properly used.
- xii. Air pressures are to be set at a suitable level for air actuated tool or equipment being used.
- xiii. Before changing or adjusting pneumatic tools, air pressure shall be turned off.
- xiv. Only trained employees shall use explosive actuated tools and the tool must also be unloaded when not in use.
- xv. Usage of such explosive actuated tools shall be avoided in case of places

where explosive/flammable vapours or gases may be present.

- xvi. Explosive actuated tools and their explosives shall be stored separately and be taken out and loaded only before the time of immediate use.
- xvii. Misfired cartridges of explosive actuated tools must be placed in a container of water and be removed safely from the project.
- xviii. No worker shall point any power operated / hand tool to any other person especially during loading / unloading.

#### **44. Dangerous and Harmful Environment**

- 44.1 When internal combustion engines are to be used into a confined space e.g. excavation, tunnel or any other workplace where neither natural or artificial ventilation system is inadequate to keep carbon monoxide below 50ppm, exposure of building workers shall be avoided unless suitable measures are taken and provided by the contractor.
- 44.2 No worker shall be allowed into any confined space, tank, trench or excavation wherein there are dusts, fumes / vapours or other impurities which is likely to be injurious or offensive, explosive, poisonous, noxious, gaseous or other harmful substance unless steps are carried out by the contractor and certified by the responsible person to be safe

#### **45. Corrosive Substances**

- 45.1 Corrosive substances including alkalis and acids shall be stored and only used by persons who have received adequate training in dealing with such substances. Suitable PPE shall be provided by the contractor to the worker during such handling and work. In case of spillage of such substances on building worker, the contractor shall take immediate remedial measures.

#### **46. Batching Plant / Casting Yard**

- 46.1 Contractor shall submit the Batching plant /Casting Yard site layout and design to the Employer / Engineer for Notice of no objection
- 46.2 The contractor shall ensure that
  - i. The batching plant / casting yard shall be effectively planned and designed for efficiency in: unloading and stacking of aggregates, steel, cement, batching plant, transport of concrete, casting the segment / girder, stacking of segments / girders, loading of segments / girders on the transporter.
  - ii. The batching plant / casting yard shall be barricaded and made as a compulsory PPE zone with restricted access provisions.
  - iii. Electrical system shall also be suitably planned so that location of diesel generator, if any, location of DBs, routing of cables and positioning of area lighting poles/masts does not infringe on any other utilities.
  - iv. Drainage shall be provided and waste water shall be disposed after treatment
  - v. Time office, canteen, drinking water, toilet and rest place shall be suitably located for the easy access to workers. All the facilities shall be properly cleaned and

maintained during the entire period of operation.

- vi. Manual handling of cement shall be avoided to a larger extent. Whenever it is absolutely necessary the workmen shall be given full body, hand and respiratory protection.
- vii. The PPE provided for cement handling shall conform to international standards.
- viii. Access roads and internal circulation roads shall be well laid and maintained at all times.
- ix. The minimum **height of stack** to be provided with each **generator set** can be worked out using the following formula:  $H = h + 0.2 \times \text{ÖKVA}$ .  $H$  = Total **height of stack** in metre.  $h$  = **Height** of the building in metres where the **generator set** is installed.
- x. DG shall be silent type with acoustic enclosure.
- xi. Non-adherence to any of the above provision shall be penalised as per relevant penalty clause.

## 47. Radiation

- 47.1 The use of radioactive substances and radiating apparatus shall comply with the Govt. regulatory requirements .
- 47.2 Operations involving ionising radiation shall only be carried out after having been reviewed without objection by the Employer / Engineer and shall be carried out in accordance with a method statement.
- 47.3 Each area containing irradiated apparatus shall have warning notices and barriers, as required by the Regulations, conspicuously posted at or near the area.
- 47.4 Radioactive substances will be stored, used or disposed shall be strictly in accordance with the Indian legal requirements.
- 47.5 The contractor shall ensure that all site personnel and members of the public are not exposed to radiation.

## 48. Smoking

- 48.1 Smoking within the working areas is prohibited with the exception of designated smoking areas near the site office. The Contractor shall provide suitably signed designated smoking areas within the near the site office close to welfare facilities but away from the work locations, site access routes and sensitive neighbouring properties. The Contractor shall take all reasonable measures to prevent personnel under his control from smoking on the streets adjacent to worksites.

## 49. Mobile Phones, Radios and Audio Equipment

- 49.1 All workers / operators are ban from using handphone/ MP3 devices at CMRL worksites. The handphones are to be surrender to respective supervisors and stored away at rest areas with lockers. Workers/ Operators are only allowed to use their handphones/ MP3 during rest times and breaks.

49.2 The Contractor shall provide suitably signed designated areas which shall include offices and welfare/rest facilities. The Contractor shall ensure that the use of mobile phones do not create risks to the user or others.

49.3 Use of personal radios and portable audio equipment is prohibited within the Working Areas with the exception of welfare/rest facilities.

## **50. Dust, gases, fumes etc.**

50.1 The Contractor shall prevent concentration of dust, gases or fumes by providing suitable means to control their concentration within the permissible limit so that they may not cause injury or pose health hazard to a building worker at a building or other construction work.

50.2 The contractor Shall look into the health hazards associated with handling different types of explosives, chemicals and other construction materials and to suggest medial measures including use of proper personal protective equipment

50.3 Provide a vacuum machine with water sprayer to wet vehicular access.

## **51. Vehicular Traffic**

51.1 Whenever any building or other construction work is being carried on, or is located in close proximity to a road or any other place where any vehicular traffic may cause danger to building workers,

(1) The Contractor shall ensure that such building or other construction work is barricaded and suitable warning signs and lights displayed or erected to prevent such danger.

(2) The Contractor shall ensure that all vehicles used at construction site of a building or other construction work comply with the requirements of the Motor Vehicles Act, 1988 (Central Act 59 of 1988).

(3) The Contractor shall ensure that a driver of a vehicle of any class or description operating at a construction site of a building or other construction work holds a valid driving licence under the Motor Vehicles Act, 1988 (Central Act 59 of 1988).

## **52. Handling of explosives**

52.1 The contractor shall ensure at a construction site of a building or other construction work that,

(a) all explosives are handled, used or stored in accordance with the instructions and the material data sheet supplied by the manufacturer of such explosives.

(b) the use of explosives is carried out in safe manner to avoid injury to any person and under the direct supervision of a responsible person; and

(c) before using any explosive, necessary warning and danger signals are created, at conspicuous places of such use to warn the building workers and the general public of the danger involved in such use.

### **52.2 Precautions:**

52.2.1 The Contractor shall ensure at a construction site of a building or other construction work that.

(a) The following precautions are observed at the places of transporting, handling, storage and use of such explosives, namely: -

- (i) prohibition of smoking, naked lights and other sources of ignition in the vicinity where explosives are handled, stored and used;
- (ii) to keep safe distance and to use non-sparking tools while opening packages containing explosives; and
- (iii) to stop the use of explosives and handling thereof while the weather conditions are not suitable for such use or handling;

All measures and precautions required to be observed for use, handling, storing or transportation of explosives under the rules framed under the Explosives Act, 1884 (Central Act 4 of 1884) are observed.

### 53. Floor Openings

53.1 Unprotected holes in the floor, deck, or roof have been responsible for a number of very serious injuries. Yet, through planning and personal attention, falls through openings under foot are very easy to prevent.

- i) If you make a hole, guard it. Before cutting the hole, barricade the work area to keep people out.
- ii) If the hole must be open, install permanent barricades around the perimeter. Like railings around the edge of a building or stairwell, these should be a minimum of forty-two inches high, have a mid-rail and be capable of withstanding a 200-pound load. Toe boards are recommended at all times and are required if anyone is going to work under the hole.
- iii) If the hole is not guarded, it must be covered. The cover must be capable of supporting at least a 200-pound load, be larger than the opening, secured against displacement, and
- iv) labelled "Floor Opening, Do Not Remove". If you have a choice, make a round opening and cover.
- v) Never cover a hole with any type of non-substantial material such as paper, cardboard or plastic.

### 54. Evaluation, Selection and Control of Sub-Contractors

54.1 The Contractor shall include legislative and site specific OHS&E requirements in tender packages for their sub-contractor selection and conduct pre-job meetings to address job OH&S expectation before awarding them.

54.2 The Employer /Engineer may participate in pre-tender meetings, selection process, pre-job meeting reviewing the OH&S performances, etc.

### 55. OH&S Sharing & Site Visits



- 55.1 The Contractor shall facilitate and host visits by other Contractors to his worksites and to share his OH&S management experiences with them.
- 55.2 The Contractor's management team shall attend visits to other CMRL worksites with good OH&S management initiatives and skills when directed by the Employer /Employer's Representative and to learn and implement the good initiatives at his site.

## **56. CCTV**

- 56.1 The Contractor shall implement a surveillance CCTV system in Administrative areas, Construction areas (general indoor, general outdoor, tunnel and general underground work areas, mucking and scaling), Maintenance / Operating areas, Mechanical/electrical equipment rooms, Warehouses and storage rooms/area, Casting yard, Labour Colony, Health Centres and First aid stations and infirmaries, Parking areas, Visitor areas and Laboratories , with cameras strategically positioned at high-risk areas for purpose of monitoring site conditions and deterring unsafe work practices. The number and location of cameras installation shall be subjected to the acceptance of the Employer /Engineer. The contractor shall submit the CCTV installation and monitoring plan to Employer /Engineer.
- 56.2 The CCTV shall facilitate viewing of live and recorded images. Access to viewing and controlling of all cameras shall be via a standard web browser and/or wireless Local area Network (LAN) by the authorized users. All cameras shall be weatherproof and come with pan/tilt functions, zoom lens and the ability to operate under low light conditions.
- 56.3 All camera recordings shall have camera id and location/area of recording as well as date/time stamp which cannot be altered, ensuring the audit trail is intact for evidential purposes. Sufficient storage (hard disk space) shall be provided for all the camera recordings for a period of 30 days or more @ 30 frames per second (FPS), at 4 common intermediate format (CIF) or better quality using the necessary compression techniques for all cameras. A backup system shall be maintained to protect against server or storage failure.
- 56.4 The storage system should allow retrieval of data instantaneously or any date/time interval chosen through search functionality of the application software. The system shall have the facility to export the desired portion of clipping (from a desired date/time to another desired date/time) onto a CD, DVD or any other device in a format which can be replayed through standard PC based software.

## **57. Barricade / Cover to Voids, Trench, Bored Holes and Open Edges of Structure**

- 57.1 The Contractor shall barricade all excavation, bored holes, voids and open edges of structure under construction where a workman is liable to fall with secured and effective guardrails / barricades / floor coverings.
- 57.2 The Contractor shall provide “Danger” warning signs for barriers and barricades erected. All floor opening covers shall be stencilled or painted with “Danger, Do Not Remove”.
- 57.3 Where traffic flow is to be maintained over temporary road opening or crossing, the Contractor shall provide suitably designed chequered steel cover / decking over it. The Contractor’s professional Engineer shall design and submit the cover / decking proposal to the Employer /Engineer for acceptance prior to commencement of the excavation.

## 58. General Safety

- 58.1 The Contractor shall provide, erect, maintain and finally remove, when instructed, Danger, Warning, Caution or Information signs, located appropriate to the site layout. The signs shall be no less than 1.5m x 1.0m in size written boldly in the three official languages. These shall be erected on existing footpaths and at points of access likely to be used by the public to warn or inform them of the existence of the Works. These notices shall be in addition to any notices required to be put up to meet statutory requirements.
- 58.2 The Contractor shall ensure that all roads, pavements and public footpaths are kept clear of dust, silt and debris.
- 58.3 Unless otherwise agreed, the Contractor shall be responsible for the proper fencing, hoarding, lighting, guarding and watching of the Works. The Contractor shall also provide proper temporary roadways, footways, guards, fencing and hoardings so far as the same may be necessary for the accommodation and protection of the owners and occupiers of the adjacent property, the public and others for a like period.
- 58.4 All platforms, covers, ladders, stairways, staging, scaffolding and other provisions for access erected by the Contractor shall be installed in compliance with current legal requirements and made available for use as early as possible during the construction period. In cases where this is impracticable the Contractor shall provide all necessary temporary access facilities which shall be constructed, installed and maintained in a safe and secure manner.
- 58.5 Designated walkways along walers and struts shall be levelled, flushed without tripping hazards and with rigid guardrails and toe boards securely provided.

- 58.6 Fall Arrest System, including but not limited to life lines and anchors, shall comply with the BS EN /IS requirements.
- 58.7 The Contractor shall provide capping on all protruding starter reinforcement bars with individual plastic/ rubber caps or with hose /tube.
- 58.8 The Contractor shall ensure bar chairs supporting steel reinforcement bars shall be designed by a Professional Engineer. In his design, the bar chair should only be welded to temporary bars incorporated to support the bar chairs. Welding onto permanent bars will not be permitted.
- 58.9 The Contractor shall ensure that horseplay, practical jokes, scuffling, wrestling or fighting are strictly prohibited at the Site.
- 58.10 The Contractor shall ensure that the sale, keeping or consumption of liquor and prohibited substances on site is prohibited.
- 58.11 The contractor shall organize community interaction programme with the general public for the awareness of the projects.
- 58.12 Provide proper and durable wire mesh overlay across rebar surface.
- 58.13 Provide light-Emitting diode (LED) lights to highlight safe access and egress in the work area.
- 58.14 Provide portable lighting detector for workers.
- 58.15 All contractors shall provide overhead protections as per Rule 41 of TBOCWR 2006
- i) Overhead protection (Fan) should be erected along the periphery of every building which is under construction and the building height shall be 15m or above after construction.
  - ii) Overhead protection shall be minimum 2m wide and the outer edge shall be 150mm higher than the inner edge and an angle not more than 200 to its horizontal sloping into the building.
  - iii) Overhead protection shall not be erected more than a height of 5m from the base of the building.
  - iv) Areas of inadvertent hazard of falling of material shall be guarded or barricaded to prevent access and appropriately sign.

## 59. Safety (Anti-Fall) Net

- 59.1 The Contractor shall provide and maintain safety net system in compliance to BS EN / IS Standards to catch persons falling whilst working in any location from where he would liable to fall. The net shall be of sufficient size and strength to catch any person for whose protection it is to be used and the net shall be so located to cover the area of the possible fall/Material.
- 59.2 The Contractor shall conduct a sample test on the safety net system, comprising the net and its supporting structures, before it is installed.
- 59.3 SAFETY CATCH NET which is utilizing at CMRL site premises should be in conforms to IS 11057-1984 standards.

## **60. Site Transport**

### **60.1 General**

- 60.1.1 All vehicles driven on site shall be maintained in roadworthy condition and be registered with the appropriate authority in accordance with the Motor Vehicles Act. Each driver of these vehicles shall hold a valid driving licence authorising him to drive that class of vehicle. Such vehicles include, but are not limited to Cement lorries; Pick-up and flat bed trucks; Lorry loader with articulating arms; and Rough terrain forklifts.
- 60.1.2 Transportation of personnel on flat bed trucks, cranes, forklifts, dumpers and similar vehicles not designed to carry passengers is prohibited.
- 60.1.3 Free-standing chairs shall not be used as seating. All passengers must be seated at all times that the vehicle is in motion and shall not have any part of their body outside the vehicle. No more persons shall be carried in the cab of any vehicle than the number for which it is licensed.
- 60.1.4 An appropriate speed limit shall be set and enforced on site.
- 60.1.5 The Contractor shall appoint banksman to control reversing vehicles and congested machinery movement. The banksman shall wear a high visibility vest clearly marked “Traffic Controller” or similar and trained in the standard hand signals and always standing to the side of the driver’s cab, not to the rear of the vehicle.
- 60.1.6 Any routes on the Site where headroom is restricted shall have appropriate warning signs posted at the approach to such restriction.
- 60.1.7 The Contractor shall ensure that the drivers of all delivery vehicles to Site and those removing materials from Site, wear appropriate PPE including, but not limited to, safety footwear and safety helmets at all times that they are outside the cabs of their vehicles.

- 60.1.8 The Contractor shall ensure that there is daily co-ordination of the movement of mobile plant and vehicles on Site to minimize the potential for an incident.
- 60.1.9 All Site access roads used by mobile plant / vehicles shall be constructed of hard standing and suitable for its intended purpose.
- 60.1.10 No unauthorised parking shall be permitted on the site anywhere other than at the designated parking area. Only cars bearing the contractor's permit shall be allowed to park there, and those of authorised visitors.
- 60.1.11 Site plant / vehicles shall be parked at designated locations only so that they do not cause obstructions to Site traffic.
- 60.1.12 Consideration shall be given at all times to the safety of the road users and gates should be positioned so that they minimise the additional risk to traffic at such locations as road junctions, bends, and etc.
- 60.1.13 A clear line of sight shall be maintained for all drivers of vehicles using the road, taking into consideration the speed limit of that road and the position of existing street furniture and trees / vegetation, to maximise the warning distance of approach.
- 60.1.14 If deemed necessary by the Employer /Engineer suitable mirrors shall be positioned at these points to enhance vision of traffic movement both on the roadway and on entering / leaving the Site.
- 60.1.15 Where the Site gates lead directly off a public street an effective method of controlling vehicles entering / leaving the Site shall be employed. This shall be manned by workers who have been trained in road safety and who are wearing high visibility vests bearing the wording "Traffic Controller". They should use recognisable hand signals and advice should be sought from the Traffic Police if necessary.
- 60.1.16 If deemed necessary by the Employer /Engineer a traffic lane may be coned off to allow for safe deceleration.
- 60.1.17 During the hours of darkness floodlighting of these Site entrances shall be provided to enhance visibility of such traffic controllers, but care must be taken that these lights do not dazzle any on-coming traffic or pedestrians or cause a nuisance to neighbouring residents. The traffic controllers should utilise hand held batons or gloves incorporating reflective material.

60.1.18 Warning signs in compliance with the IRC standards shall be conspicuously displayed at appropriate distances before such gates into the Site as to give all drivers a clear understanding of the traffic hazard ahead

60.1.19 If vehicles entering / leaving the site have to cross a public footpath or pavement then a worker must be deployed to control pedestrians as well as road traffic. Suitable warning signs should be deployed to alert pedestrians to possible traffic movement across the footpath.

## **60.2 Spillages On Roads**

60.2.1 A paved truck wash bay for washing vehicles leaving the worksite onto a roadway shall be provided and maintained at each vehicular egress point before commencement of works on Site. Each truck wash bay design must be approved by the Employer /Engineer before it can be constructed on site.

60.2.2 Wash water from the wash bays shall be directed into a water treatment plant for treatment.

60.2.3 Preventative measures shall be taken to limit the incidence of earth droppings from earth moving vehicles. In the event that any earth dropping occurs onto a public road/drain, such earth shall be removed and the roads / drains washed by the contractor at his own expense to the satisfaction of the Employer /Engineer.

60.2.4 Contractor shall assign personnel and establish a system of checks to ensure that all vehicles and trucks leaving the worksite do not have the potential to litter the roads due to its wheels or transportation materials.

60.2.5 Where it is foreseeable that water may drain out from a load of soft marine clay or similar transported wet materials that will contaminate the road surface, the Contractor shall ensure that specially designed and constructed watertight trucks are used to transport these materials.

60.2.6 All cement mixer trucks servicing CMRL sites must have a containment system or a flap installed to prevent spillage of cement.

60.2.7 Every vehicle or earth moving equipment shall have the following fully functional: Silencers, Taillights , power and hand brakes , reversing alarm and search light forward and backward.

60.2.8 In case of non-conformity of items identified by the Employer/Engineer shall take necessary steps including stoppage of work and or imposing of any penalty to ensure satisfactory execution.

## **61. Site Security**

### **61.1 General**

- 61.1.1 The Contractor is responsible for the security of the sites, works areas, material storage areas, site offices, facilities and the Works.
- 61.1.2 Security for the site and the works shall be maintained throughout the duration of the contract or the extended period as provided in the particular specification.
- 61.1.3 The Contractor is to provide a detailed security plan which includes security proposals for the various phases of the construction of the works.
- 61.1.4 The security for the various phases of the works shall include the set up phase, temporary works phase, structural works phase, E&M works and architectural works phase as well as the completion phase.
- 61.1.5 For each phase, the contractor shall detail the security measures, facilities, guards and patrols to be implemented.
- 61.1.6 Within 28 days the contractor shall submit a fully detailed security Plan detailing the contractor's propose security measures and facilities he intends to implement throughout the contract duration. The security plan shall include but not limited to the followings:
- (a) Method of securing all site and works areas; offices and facilities, etc.;
  - (b) The security facilities to be set up and its locations;
  - (c) Lightings, alarms, communication equipment's, cameras, surveillance equipment's, etc.;
  - (d) Layout and site plans;
  - (e) Personnel, manpower and the organisation chart;
  - (f) Guards and patrols, numbers, locations and frequency;
  - (g) Security for various phases or stages of the Works;
  - (h) Controls to be implemented for access by authorised personnel/ contractors/ Construction Equipment/ Plant/ vehicles/ materials;
  - (i) Control of access points for visitors and their vehicles entering the Site, issue of entry permits and maintaining records for every visitor and vehicle into the Site, including their stated purpose;
  - (j) Monthly review and reports;
  - (k) Incident report and review;
  - (l) Audits; and
  - (m) Others.
- 61.1.7 The plan shall include a central security post/ office manned at all times by a competent uniformed supervising security guard and an assistant from a licensed Security Agency at an agreed location. It shall be possible to communicate between each of the security posts/office and the central security post/office.
- 61.1.8 The Contractor shall update the Security Plan regularly and when the Works enters a new phase to ensure that the Site, facilities, and Works are adequately and sufficiently



protected against theft, wilful damages, misdemeanours, and other illegal or undesirable activities. Sufficient deterrence shall be implemented to be in tandem with the progress of the Works.

61.1.9 The Contractor shall be aware that there may be other contractors employed or engaged by the Authority to carry out works within the Site. Security measures shall cover these works carried out by other contractors, their site storage and facilities.

61.1.10 Security audit shall be carried at 3 monthly intervals to detect lapses and other inadequacies of the Security Plan. Audit reports and proposed corrective actions shall be submitted to the Employer /Engineer.

61.1.11 The Contractor shall improve his site security and or implement other measures required by the Employer /Engineer when he is of the opinion that the security measure is insufficient or where there are lapses in the security system.

## **61.2 Hoardings**

61.2.1 The safety of vehicular movement onto and off site shall be planned before the site hoardings are erected to allow for features to be incorporated maximising public safety in connection with site activities.

61.2.2 The hoarding shall be well designed and secured in place to prevent it being blown over by gusts of wind and shall be sufficiently robust to deter anyone from removing or displacing any panels. Regular maintenance shall be carried out.

61.2.3 At all works areas; suitable and effective gates shall be provided and shall be locked during non-working hours/days. Guard posts shall be constructed and guards shall be stationed at these posts. It shall be illuminated to enable the guards to carry out checking at night. The guard shall check persons, vehicles, materials and other equipment entering and leaving site. Intercom or other communications shall be provided and maintained between the guard post and the security guard office.

61.2.4 At strategic locations along hoardings, entrances, guard posts and other facilities, spot lights shall be installed.

61.2.5 There shall be no unauthorised openings in the perimeter hoarding. All access and egress shall be via gates which are manned by 24-hour security guards.

61.2.6 If for any reason the contractor authorises any panels to be moved to create a temporary opening then a guard must be posted to restrict access to authorised personnel/ construction equipment/ plant/ vehicles only. It shall permanently close immediately after works are completed.

61.2.7 Should there be any risk of materials, tools, waste material or similar escaping from any part of the site either as projectiles or falling objects etc., then an effective extension for the Site hoarding shall be designed and erected to prevent this. Where a public footpath or pavement runs alongside the hoarding then overhead protection must be provided where site activities pose a foreseeable risk of such occurrences.

### 61.3 Personnel Access

- 61.3.1 The contractor shall be responsible for controlling worksite security to prevent unauthorised access, maintain public safety and other offences.
- 61.3.2 The contractor shall ensure that no illegal workers or unauthorised persons are allowed on Site. The contractor shall conduct both internal and external raids to weed out any illegal workers or unauthorised persons.
- 61.3.3 Security of the site shall also be maintained to ensure that only those persons who have the skills and training to work safely on site and who are wearing the appropriate personal protective equipment (PPE) may be admitted.
- 61.3.4 At the main Site office, the contractor shall set up a fully equipped security guard office at the entrance to the offices. The security guard office shall be at least 6m x 3m with direct view overlooking the entrance. A visitor book shall be maintained to record visitors entering/leaving the site or offices. Workers' security passes shall also be issued to all workers entering the Site. Visitor shall be issued with temporary passes. Ingress and egress of vehicles shall be logged.
- 61.3.5 A similar security pass system shall also be operated at the main entrance to all the sites. This shall preferably incorporate a turnstile / swipe card reader but any other equally effective means may be utilised with the prior acceptance of the Employer /Engineer.
- 61.3.6 Guard posts erected at the main entrances where access is to be so controlled are to be staffed by trained employees of the Contractor or employees of a subcontract licence security agency. Sufficient guards shall be on duty at any one time to give effective 24-hour cover. These trained guards shall wear uniform to enable easy identifications.
- 61.3.7 All guard posts shall be equipped with a telephone / radio communication system, a panic button and an audible alarm.
- 61.3.8 Two-way communication equipment shall be provided to the guards to maintain communication between guards at other security posts or on patrol and the security guard office. A general alarm system shall be installed for use in emergency. The Employer /Engineer site offices shall be installed with intruder alarms, and protection systems. The site offices surrounding shall be well lit
- 61.3.9 The access control system shall cover all staff, direct employees of the contractor, Interfacing contractors and service providers such as canteen workers, cleaning workers and similar as well as all subcontract staff / workers and those of sub-subcontractors etc. including operators of rental construction equipment / plant.
- 61.3.10 Each security pass shall bear the photograph of the holder and his registration number as appropriate. It shall be issued after completion of site safety induction training via a central registry controlled by the contractor. Temporary visitor passes may be issued

for those personnel on authorised business, in which case their names must be recorded together with times of arrival and departure plus signature.

61.3.11 The contractor shall issue first time security passes to the Interfacing contractors. The cost of subsequent replacement of security passes will be borne by the Interfacing contractors.

61.3.12 Persons not wearing the correct PPE shall not be allowed onto site, unless they identify themselves as visiting the site office only and the Contractor has provided a designated safe access route to and from the site access control point for this purpose.

61.3.13 An accurate headcount shall be kept of all persons entering the worksite so that they can all be accounted for in case of an emergency.

#### **61.4 Vehicular Access**

61.4.1 Effective control shall be exercised over materials entering and leaving the site, to check on the suitability / safety of construction equipment, plant and materials delivered to site and to prevent theft. For all materials including waste and salvaged materials, construction equipment and plants leaving the site a proper record of authorisations given by the respective contractors issuing such removal chits shall be maintained.

61.4.2 Wherever practicable separate access and egress gates shall be provided. The control point for access should be located a short distance inside the main gate to permit a vehicle to pull off the road before halting to be checked.

61.4.3 A security post shall be provided at each vehicular access point, sited so that the checker has a clear view of incoming vehicle registration plates and any passengers in the cabin.

61.4.4 The Contractor shall provide the following at all designated site entrances and exits for vehicular access: -

- i) Illumination of at least 100lux;
- ii) Convex mirror (32 inch/ 24 inch) shall be provided at entrances/exit point adjoining roads; and
- iii) Revolving light (complying to BS3143 Part 4 or an approved equivalent).
- iv) Vehicle lay-by at all site entrances/ exits point where possible.

61.4.5 Railings or similar should be erected inside the site to prevent workers and others on site from walking into the path of vehicles entering / leaving the Site.

61.4.6 To prevent all lifting machines and vehicles from infringing the height limit of 4.5m in public roads. The contractor shall install an effective and robust steel height barriers at ALL vehicular access. The steel height barrier shall be constructed with a height infringement gauge that limits the height of lifting machines and vehicles to less than 4.5m

61.4.7 Security drop-bars shall be maintained in the closed position at all vehicular entry and exit points to the Site and every vehicle shall be stopped outside the barrier. The barrier should only be raised after authority has been given to the driver of the vehicle to proceed. Hump shall also be constructed at exit.

61.4.8 The registration numbers of all vehicles entering and leaving the site shall be recorded.

61.4.9 No goods shall be off-loaded at the site entrance in a manner that creates a hazard to other vehicles entering / leaving the site.

61.4.10 Vehicles waiting to get onto the site should be directed to a designated holding area which minimises obstruction to other road users, and called forward by a flagman.

61.4.11 The main gates shall be closed and locked after construction work has ended for the day.

61.4.12 The main gates and control points are to be well lit during the hours of darkness.

## **61.5 Site Patrols**

61.5.1 Security guards shall patrol the site regularly during the night, non working days and public holidays covering all locations within the site perimeter to deter, detect and follow up any undesirable event such as theft, robbery, violence, damage to any property, trespass, etc. The areas to be covered shall include, but not be limited to Site offices, labour colony, canteen area, materials stockyards and perimeter hoarding.

61.5.2 Clocking points shall be installed to record their presence. Guard patrols shall visit designated clocking points within the station and the perimeter of the station site at least 4 times during the night and extra 5 times during the day on non working day. A register is to be kept to record the guards visits.

61.5.3 In addition to the above when the works are near completion, guards shall patrol inside the station and clock at pre-determined clocking points at fixed intervals. Only 1 or 2 access points into the station shall be used and manned. All persons entering or leaving the station shall sign in or out and record their works or purpose. All materials and equipment (including those belonging to the Interfacing Contractors) brought into or out of the station shall be recorded. No materials or equipment shall be taken out of the station without authorisation.

61.5.4 All incidents shall be recorded and reported to the Employer /Engineer. In cases of emergency, the correct persons or appropriate emergency services shall be notified.

## **61.6 Security Guards**

- 61.6.1 Guards shall wear uniform so that they can be readily identified. They shall be able bodied, adequately trained, approved by relevant authorities and shall have no criminal records. Guards on patrol shall work in pairs.
- 61.6.2 Personal Protection Equipment (PPE) shall be provided to the guards by the Contractor and they shall wear them when patrolling or visiting the sites areas. They shall also attend the safety induction course.
- 61.6.3 Guards shall be fully trained to handle various situations such as unauthorized site entry, theft, fire, accident, etc.
- 61.6.4 A full-time guard supervisor/superintendent shall be assigned to supervise and check on the guards. He shall be trained and well versed in security procedures, measures and system, including preparing monthly reports, incident reports, reviews and audits. He shall conduct security briefing, site security campaign, provide information to workers on security and other preventive measure or deterrents.

## **62. Flooding**

### **62.1 General**

- 62.1.1 The Contractor shall consider the risks of flooding due to nature or arising from the construction activities.
- 62.1.2 The Contractor shall submit, within two (2) months of the award of the Contract a comprehensive flood protection plan for the whole of his Works. In preparing this plan, he shall carefully examine the surrounding topography to determine the probable nature and extent of any flooding. Where relevant the Contractor shall also consider in his plan the possible adverse effects of wind and tides and the risk from adjacent tunnel contracts.
- 62.1.3 The plan shall identify the areas that are at risk from flooding. The plan shall examine the potential consequences of any flooding and shall make proposals to prevent flooding.
- 62.1.4 The flood protection measures provided shall comply with the requirements of National and State legal requirements.
- 62.1.5 When pumping is proposed, the Contractor shall address the following:
- operating procedures
  - provision of a secure power supply and back-up automation and/or reaction time required to
  - start the pumps

- method and arrangement of discharge 1.6 Shafts shall be surrounded by topography sloping away from shaft to direct rain water to a drainage channel. Shafts shall be equipped with a minimum of 2 dewatering lines per shaft to ensure effective drainage in case of blockage.

## 62.2 Flooding Evacuation and Rescue Procedure

62.2.1 The Contractor must make adequate safety provisions in the event of flooding whether it is due to inclement weather or the river bank bursting in order to minimise the loss of life and property damage, where practicable.

62.2.2 The safety provisions to be put in place shall not be limited to providing of an emergency evacuation and rescue plan, trained rescue teams and water rescue aid or equipment. The trained rescue team and emergency procedure shall be in place before the start of any work. The number of trained rescue teams and water rescue aid or equipment shall be increase when deem necessary by the Employer /Engineer. The water rescue aids or equipment shall be properly maintained throughout the lifetime of the project.

62.2.3 Emergency drills shall be conducted every month with the involvement of State Govt rescue team where necessary.

62.2.4 The following are the minimum requirement of Emergency Equipment shall be available at site but not limited too.

### S.No Equipment & Minimum qty

1. Motor Trolleys – 2 No.
2. Dip Lorry – 1 No.
3. Re-railing vehicle and Re-railing Equipment – 1 Set
4. Gas cutting equipment – 1 No.
5. Search Lights – 2 No.
6. Track continuity Jumpers – 2 No.
7. Fire Extinguishers – 3 No.
8. Hand Signals / Flags – 1 No.
9. Crank handle key, Clamps and Pad locks – 1 No.
10. First Aid Kits – 1 No. per station
11. Stretchers – 1 No. per station
12. Wheel Chair – 2 No. per station
13. Pick Axe - 2 No,
14. Self Contained Breathing Equipment – 2 No.
15. Inflatable tents – 1 No.
16. Collapsible lighting tower – 1 No.

17. Inflatable Air Bags – 2 No.
18. Automatic Spring loader – 1 No.
19. Track measuring gauges – 1 No.
20. Augmented First Aid Boxes – 2 No.
21. Wrist bands – As required
22. High visibility vests – As required
23. High Power torches – 2 No.
24. Nylon slings of 40 T capacity – 1 Set
25. Video Cameras – 1 No.
26. Tetra Sets, as required
27. Pumps (as per site requirements)
  - 25HP-03Nos
  - 10HP-05 Nos
  - 05 HP-05 Nos
  - 03HP-05 Nos

### **63.OH&S Submittals to the Employer/Engineer**

63.1 The contractor's OH&S management shall send the following reports to the Employer / Engineer:

- i) Daily Reporting of total no of workmen and Daily Dairy of CSM
- ii) Monthly OH&S Report
- iii) OH&S Committee Meeting Minutes
- iv) OH&S Inspection Reports
- v) Monthly Audit Rating Score (MARS) report
- vi) External OH&S Audit
- vii) Electrical Safety Audit
- viii) Monthly Tunnel inspection Checklist
- ix) Working at height safety Audit
- x) Plant and Equipment Safety Audit
- xi) Labour colony safety audit
- xii) Air and Noise Quality monitoring report

63.2 Daily Reporting of total no of workmen

63.2.1 The contractor shall report to the Employer the total no of workmen engaged including any subcontractor within 2 hours of starting of any shift in any day. This reporting shall be the primary duty of the Chief OHS&E Manager of the contractor and reporting shall be through tele-fax / email. If the information is not received or received more than 2 hrs after starting of the shift, penalty shall be levied as per relevant clause.

63.3 The Chief OHS&E Manager shall prepare and submit the daily dairy (FSAF 29) to the Employer/Engineer on the end of each shift.



## **64. Monthly OH&S Report**

64.1 The contractor shall prepare a monthly OH&S report are given in general instruction GSAF 14 and submit by the 7th of every month to the Employer/Engineer.

## **65. Visitors to Site**

### **65.1 General**

65.1.1 All visitors to site shall report to the Contractors site offices where they shall be issued with appropriate Personal Protective Equipment if they are to go out onto the site work areas. Any visitors going out to the site work areas shall be accompanied at all times by a member of the site personnel.

## Appendix

The Contractor shall also comply with the following Section 1 & 2 to this appendix.

Section 1 –Standard Forms

Section 2 –Guidance Documents

### SECTION 1

#### STANDARD FORMS

| INDEX                                          | REFERENCE | REVISION | DATE |
|------------------------------------------------|-----------|----------|------|
| Pre-Task Check Form                            | FSAF 01   |          |      |
| General Permit                                 | FSAF 02   |          |      |
| Hazard Log & Risk Register                     | FSAF 03   |          |      |
| Scaffold inspection register                   | FSAF 04   |          |      |
| scaffolding Handover certificate               | FSAF 05   |          |      |
| Crane Operator Weekly Inspection Check list    | FSAF 06   |          |      |
| Excavation Permit                              | FSAF 07   |          |      |
| Hot Work Permit                                | FSAF 08   |          |      |
| Near miss report                               | FSAF 09   |          |      |
| Confined Space                                 | FSAF 10   |          |      |
| MARS Score Sheet                               | FSAF 11   |          |      |
| First aid slip                                 | FSAF 12   |          |      |
| CMRL Template Site Inspection Form             | FSAF 13   |          |      |
| Record of Induction Training                   | FSAF 14   |          |      |
| Induction Checklist                            | FSAF 15   |          |      |
| Operations Induction                           | FSAF 16   |          |      |
| Visitors Induction                             | FSAF 17   |          |      |
| MEWP Checklist                                 | FSAF 18   |          |      |
| Permit to work on electrical                   | FSAF 19   |          |      |
| Permit to excavate                             | FSAF 20   |          |      |
| Excavator Checklist                            | FSAF 21   |          |      |
| Oil storage Requirements                       | FSAF 22   |          |      |
| Harness Checklist                              | FSAF 23   |          |      |
| Monthly contractor's Accident Statistic Report | FSAF 24   |          |      |
| Formation of Site OHS&E Committee              | FSAF 25   |          |      |
| Minutes of OHS&E Committee Meeting             | FSAF 26   |          |      |
| Electrical Safety Audit Inspection Checklist   | FSAF 27   |          |      |
| Contractor Monthly Inspection Checklist        | FSAF 28   |          |      |
| Chief OHS&E Manager Daily Dairy                | FSAF 29   |          |      |
| CV Template                                    | FSAF 30   |          |      |
| External Quarterly Audit Template              | FSAF 31   |          |      |

### PRE-TASK CHECK

|                             |            |
|-----------------------------|------------|
| Site:                       | Date:      |
| <b>TASK &amp; LOCATION:</b> |            |
| Supervisor Name (PRINT):    | Signature: |

| <u>Item</u> | <u>Control Measures</u>                                                                                                                                                                     | YES✓ or NO X |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| 1           | Is all equipment and correct tools available to undertake the task and in correct working order? E.g. crane, Rig, Hydra, JCB certification, launch equipment scaffold etc.                  |              |
| 2           | Are men trained and competent to undertake the task including tools and equipment? E.g. crane / Rig operators, welders and certification available                                          |              |
| 3           | Is all safety equipment available to do the task e.g. head, hand, eye, hearing protection, harness, concrete burn protection? Are men trained how to use it? E.g., harness fitting and use. |              |
| 4           | Have all men been briefed and instructed on the specific task to be undertaken by the task supervisor prior to commencing operations?                                                       |              |
| 5           | Are all safety precautions in place as identified in the risk assessment & method statement e.g. edge protection, access platforms, cordon & barriers etc?                                  |              |
| 6           | Have all men been briefed on the safety precautions, procedures and controls that must be followed as per the method and risk assessment?                                                   |              |
| 7           | Is competent supervision in place? E.g. suitable knowledge, training & experience in specific task to be supervised                                                                         |              |

***IF THE ANSWER IS 'NO' WORK CANNOT START !!!!!!!***



## GENERAL PERMIT

Number.....

General Permit can be used where control of an operation is required to ensure safety to public and or operatives.

The type of operation that this permit could be used for includes:

Any high risk areas identified in the risk assessments

e.g. Work in or near public spaces, demolition

### PART A – ISSUE

Permit is issued to: .....

of: .....

for (specific work to be done) .....

.....

.....  
at (exact location of work) .....

.....

Valid from: ..... am/pm

to: ..... am/pm Date: .....20....

**I issue this permit subject to the minimum precautions  
I have listed in Part B below.**

**Signed:** .....  
(Issuer)

### PART B - PRECAUTIONS

- Liaise with occupiers ☐
- Post warning notices ☐
- Provide suitable barriers ☐
- Adequate competent supervision ☐
- Other permits e.g. Confined Space ☐
- Provide suitable traffic control ☐
- Location protected and sign marked ☐
- Other ..... ☐
- ..... ☐
- ..... ☐
- ..... ☐

### PART C - ACCEPTANCE

**I accept this permit and will ensure the precautions  
ticked above in part B have been undertaken before**

### PART D - COMPLETION OF WORK

The work has been completed and the work area  
and adjacent areas have been thoroughly checked  
to ensure no hazards remain

- ☐  
All equipment and materials removed
- ☐  
Safety equipment returned
- ☐  
Area restored
- Other

.....

.....

..... ☐

**The work has been completed, and I have undertaken  
the checks and safety measures I have ticked in Part  
D of this permit above.**

Time: .....am/pm

Date: .....20.....

**Signed:** .....  
(subcontractor once completed work)

### PART E - CANCELLATION OF PERMIT

**This permit to work is cancelled and the relevant  
staff have been notified. No more work under this  
permit.**

Time: .....am/pm

Date: .....20.....

**Signed:** .....  
(Issuer)



starting work. I also understand the checks and safety measures I am required to undertake after completion of work.

Signed: ..... (Subcontractor undertaking work)



## Hazard Identification and Risk register

| Identified Hazard          | Applicable | Risk Rating |   |   | Location<br><br>(List Where Necessary) | Method Statements<br><br>Risk Assessment<br>Required | Minimum Mitigation<br><br>(Details to be Completed if Box ticked)<br><br>(E.G. Procedures to be followed) | Procedure<br>Reference<br>Number |
|----------------------------|------------|-------------|---|---|----------------------------------------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|----------------------------------|
|                            |            | H           | M | L |                                        |                                                      |                                                                                                           |                                  |
| Site (External)            |            |             |   |   |                                        |                                                      |                                                                                                           |                                  |
| Access and Egress          |            |             |   |   |                                        |                                                      |                                                                                                           |                                  |
| Existing roads             |            |             |   |   |                                        |                                                      |                                                                                                           |                                  |
| Contaminated ground        |            |             |   |   |                                        |                                                      |                                                                                                           |                                  |
| Unstable ground            |            |             |   |   |                                        |                                                      |                                                                                                           |                                  |
| Ground<br>water/conditions |            |             |   |   |                                        |                                                      |                                                                                                           |                                  |
| Underground services       |            |             |   |   |                                        |                                                      |                                                                                                           |                                  |
| Overhead services          |            |             |   |   |                                        |                                                      |                                                                                                           |                                  |
| Existing traffic           |            |             |   |   |                                        |                                                      |                                                                                                           |                                  |
| Traffic restrictions       |            |             |   |   |                                        |                                                      |                                                                                                           |                                  |
| Existing plant or works    |            |             |   |   |                                        |                                                      |                                                                                                           |                                  |
| Residential buildings      |            |             |   |   |                                        |                                                      |                                                                                                           |                                  |
| Hotels                     |            |             |   |   |                                        |                                                      |                                                                                                           |                                  |

|                      |  |  |  |  |  |  |  |  |
|----------------------|--|--|--|--|--|--|--|--|
| Industrial buildings |  |  |  |  |  |  |  |  |
| Schools/hospital     |  |  |  |  |  |  |  |  |
| Existing noise       |  |  |  |  |  |  |  |  |
| Existing dust        |  |  |  |  |  |  |  |  |
| Existing fumes       |  |  |  |  |  |  |  |  |
| OTHERS List: -       |  |  |  |  |  |  |  |  |



| Construction Operations: |            |   |   |   |                 |       |                    |                            |
|--------------------------|------------|---|---|---|-----------------|-------|--------------------|----------------------------|
| Identified Hazard        | Applicable | H | M | L | Location (List) | MS/RA | Minimum Mitigation | Procedure Reference Number |
| Scaffolding              |            |   |   |   |                 |       |                    |                            |
| Working at height        |            |   |   |   |                 |       |                    |                            |
| Steel erection           |            |   |   |   |                 |       |                    |                            |
| Confined spaces          |            |   |   |   |                 |       |                    |                            |
| Temporary works          |            |   |   |   |                 |       |                    |                            |
| Site clearance           |            |   |   |   |                 |       |                    |                            |
| Bulk muck shift          |            |   |   |   |                 |       |                    |                            |
| Piling                   |            |   |   |   |                 |       |                    |                            |
| Girder Launch            |            |   |   |   |                 |       |                    |                            |

|                                               |  |  |  |  |  |  |  |  |
|-----------------------------------------------|--|--|--|--|--|--|--|--|
| Concrete operations                           |  |  |  |  |  |  |  |  |
| Piers / columns                               |  |  |  |  |  |  |  |  |
| Excavations                                   |  |  |  |  |  |  |  |  |
| Station Roofing Activities                    |  |  |  |  |  |  |  |  |
| Brickwork Block work                          |  |  |  |  |  |  |  |  |
| Cladding, glazing                             |  |  |  |  |  |  |  |  |
| Aluminium installation                        |  |  |  |  |  |  |  |  |
| M & E                                         |  |  |  |  |  |  |  |  |
| Electrical works                              |  |  |  |  |  |  |  |  |
| Electrical works to site distribution         |  |  |  |  |  |  |  |  |
| Plumbing inc, drainage, pipework installation |  |  |  |  |  |  |  |  |
| Fan coil installation                         |  |  |  |  |  |  |  |  |
| Air handling units                            |  |  |  |  |  |  |  |  |

|                                  |  |  |  |  |  |  |  |  |
|----------------------------------|--|--|--|--|--|--|--|--|
| Site Hoarding                    |  |  |  |  |  |  |  |  |
| Elevator installation            |  |  |  |  |  |  |  |  |
| Escalator installation           |  |  |  |  |  |  |  |  |
| Surfacing                        |  |  |  |  |  |  |  |  |
| Safety impact barrier erection   |  |  |  |  |  |  |  |  |
| Kerb Laying                      |  |  |  |  |  |  |  |  |
| Plant and equipment              |  |  |  |  |  |  |  |  |
| Lifting Operations               |  |  |  |  |  |  |  |  |
| Lifting Appliances and equipment |  |  |  |  |  |  |  |  |
| Demolition                       |  |  |  |  |  |  |  |  |
| Plant movement                   |  |  |  |  |  |  |  |  |
| Vehicle movements                |  |  |  |  |  |  |  |  |
| Piling                           |  |  |  |  |  |  |  |  |

|                                |  |  |  |  |  |  |  |  |
|--------------------------------|--|--|--|--|--|--|--|--|
| Explosives                     |  |  |  |  |  |  |  |  |
| Waste Disposal<br>Housekeeping |  |  |  |  |  |  |  |  |
| Welding                        |  |  |  |  |  |  |  |  |
| Cutting/burning                |  |  |  |  |  |  |  |  |
| Cartridge Tools                |  |  |  |  |  |  |  |  |
| Trial Holes                    |  |  |  |  |  |  |  |  |
| Others List: -                 |  |  |  |  |  |  |  |  |

| Risks to public    |            |   |   |   |                 |       |                    |                            |
|--------------------|------------|---|---|---|-----------------|-------|--------------------|----------------------------|
| Identified Hazard  | Applicable | H | M | L | Location (List) | MS/RA | Minimum Mitigation | Procedure Reference Number |
| Access to site     |            |   |   |   |                 |       |                    |                            |
| Movements of plant |            |   |   |   |                 |       |                    |                            |
| Live traffic       |            |   |   |   |                 |       |                    |                            |

|                      |  |  |  |  |  |  |  |  |
|----------------------|--|--|--|--|--|--|--|--|
| Noise/dust           |  |  |  |  |  |  |  |  |
| Parking              |  |  |  |  |  |  |  |  |
| Fire                 |  |  |  |  |  |  |  |  |
| Storage of materials |  |  |  |  |  |  |  |  |
| Falling materials    |  |  |  |  |  |  |  |  |
| OTHERS List:-        |  |  |  |  |  |  |  |  |

| Health Risks      |            |   |   |   |                 |       |                    |                            |
|-------------------|------------|---|---|---|-----------------|-------|--------------------|----------------------------|
| Identified Hazard | Applicable | H | M | L | Location (List) | MS/RA | Minimum Mitigation | Procedure Reference Number |
| Weil's disease    |            |   |   |   |                 |       |                    |                            |
| Gas oil           |            |   |   |   |                 |       |                    |                            |

|                         |            |   |   |   |                 |       |                    |                            |
|-------------------------|------------|---|---|---|-----------------|-------|--------------------|----------------------------|
| Dusts                   |            |   |   |   |                 |       |                    |                            |
| Vapours                 |            |   |   |   |                 |       |                    |                            |
| Cement                  |            |   |   |   |                 |       |                    |                            |
| Bitumen                 |            |   |   |   |                 |       |                    |                            |
| Epoxy                   |            |   |   |   |                 |       |                    |                            |
| Night Working           |            |   |   |   |                 |       |                    |                            |
| Chlorine Storage        |            |   |   |   |                 |       |                    |                            |
| OTHERS List:-           |            |   |   |   |                 |       |                    |                            |
| TEMPORARY ACCOMMODATION |            |   |   |   |                 |       |                    |                            |
| Identified Hazard       | Applicable | H | M | L | Location (List) | MS/RA | Minimum Mitigation | Procedure Reference Number |
| Office installation     |            |   |   |   |                 |       |                    |                            |
| Electrical installation |            |   |   |   |                 |       |                    |                            |
| Drainage                |            |   |   |   |                 |       |                    |                            |

|                                         |  |  |  |  |  |  |  |  |
|-----------------------------------------|--|--|--|--|--|--|--|--|
| Other services                          |  |  |  |  |  |  |  |  |
| Fire                                    |  |  |  |  |  |  |  |  |
| Signing                                 |  |  |  |  |  |  |  |  |
| Cleaning                                |  |  |  |  |  |  |  |  |
| Fencing                                 |  |  |  |  |  |  |  |  |
| Security                                |  |  |  |  |  |  |  |  |
| Material storage                        |  |  |  |  |  |  |  |  |
| Sleeping accommodation and labour camps |  |  |  |  |  |  |  |  |
| OTHERS List:-                           |  |  |  |  |  |  |  |  |

| Site Specific Risks (including Designer's Risks) |            |   |   |   |                 |       |                    |                            |
|--------------------------------------------------|------------|---|---|---|-----------------|-------|--------------------|----------------------------|
| Identified Hazard                                | Applicable | H | M | L | Location (List) | MS/RA | Minimum Mitigation | Procedure Reference Number |



|  |  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|--|

| Risk Rating | Tolerance                                                                                                                                                                                                                                |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HIGH        | High severity and High probability of major injury, disablement or possible fatality. Work must cease immediately and additional control measures implemented prior to work commencement.                                                |
| MEDIUM      | Probability that an incident will occur over time (Over 3 day injury) but not as serious as a major injury or fatality. Additional control measures to be implemented within 24 Hrs.                                                     |
| LOW         | Negligible risk of injury, but task requires additional control measures to reduce likelihood of injury (Minor Injury not above 3 days off from employed role) Additional control measures to be specified and implemented within 48 Hrs |

| Severity<br>Likelihood          | Minor Injury<br>First Aid Case | Over 3 day<br>injury | Critical<br>Fatality or Major Injury |
|---------------------------------|--------------------------------|----------------------|--------------------------------------|
| Probable                        | High                           | High                 | High                                 |
| Possible in time                | Medium                         | Medium               | High                                 |
| Negligible<br>Unlikely to occur | Low                            | Medium               | Medium                               |

## SCAFFOLD INSPECTION REPORT

### NOTES

#### **General**

1. The inspection report should be completed before the end of the relevant working period (every 7 Days)
2. The person who prepares the report should, within 24 hours, provide either the report or a copy to the person on whose behalf the inspection was carried out. Eg Safety Manager, Engineer etc
3. The report should be kept on site until work is complete.

#### **Working Platforms Only**

1. Any employer or any other person who controls the activities of persons using a scaffold shall ensure that it is stable and of sound construction and that the relevant safeguards are in place before his employees or persons under his control first use the scaffold.
2. No report is required following the inspection of any mobile tower scaffold which remains in the same place for less than 7 days.
3. Where an inspection of a working platform or part thereof or any personal suspension equipment is carried out:
  - i. before it is taken into use for the first time; or
  - ii. after any substantial addition, dismantling or other alteration;

#### **Checklist of typical scaffolding faults:**

| Footings        | Standards              | Ledgers            | Bracing        | Putlogs & transoms | Couplings     | Bridles           | Ties         | Boarding    | Guard-rails & toe-boards | Ladders             |
|-----------------|------------------------|--------------------|----------------|--------------------|---------------|-------------------|--------------|-------------|--------------------------|---------------------|
| Soft and uneven | Not plumb              | Not level          | Some missing   | Wrongly spaced     | Wrong fitting | Wrong spacing     | Some missing | Bad boards  | Wrong height             | Damaged             |
| No base plates  | Jointed at same height | Joints in same bay | Loose          | Loose              | Loose         | Wrong couplings   | Loose        | Trap boards | Loose                    | Insufficient length |
| No sole plates  | Wrong spacing          | Loose              | Wrong fittings | Wrongly supported  | Damaged       | No check couplers | Not enough   | Incomplete  | Some missing             | Not tied            |



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|            |         |         |   |   |                   |   |   |                       |              |           |
|------------|---------|---------|---|---|-------------------|---|---|-----------------------|--------------|-----------|
| Undermined | Damaged | Damaged | - | - | No check couplers | - | - | Insufficient supports | Brick guards | No Ladder |
|------------|---------|---------|---|---|-------------------|---|---|-----------------------|--------------|-----------|

**HANDING-OVER CERTIFICATE – SCAFFOLDING**

Date: \_\_\_\_\_ Site: \_\_\_\_\_

Time: \_\_\_\_\_

Description &amp; location of section handed over:

---

---

Drawing No: (Where applicable) \_\_\_\_\_

Scaffolding as described above has now been completed and complies with Construction Codes of Practice. It is structurally sound and should be used and loaded in accordance with the stated specifications within this certificate.

(a) **Use only for** (General, Light, Heavy, Access etc) \_\_\_\_\_(b) **Working lifts, with distributed Load of:** \_\_\_\_\_ (kn/m<sup>2</sup>) per lift.

The detailed requirements of the Codes of Practice with regard to handrails – working platform – toe board – bracing – and ties have been complied with. This scaffold must be inspected once a week (or following exposure to wither conditions likely to have affected its strength or stability) by the user and the inspection recorded in a scaffold register. This inspection is undertaken to ensure that the scaffold continues to comply with the Codes of Practice.

N.B.Tarpauline sheets (or other wind sails) must not be fixed to a scaffold unless has been specifically designed to take them.

Scaffold Foreman: \_\_\_\_\_ Signature: \_\_\_\_\_

(date: \_\_\_\_\_)

**DO NOT REMOVE TIES**

| <b>CRANE OPERATOR WEEKLY INSPECTION CHECKLIST</b>                                                                              |   |                                         |     |                               |   |                     |     |
|--------------------------------------------------------------------------------------------------------------------------------|---|-----------------------------------------|-----|-------------------------------|---|---------------------|-----|
| Crane name/number                                                                                                              |   | Crane type:                             |     | Crane capacity                |   | Date of inspection: |     |
| Location:                                                                                                                      |   | Hour Meter: Start: _____<br>Stop: _____ |     | Total hours operated:         |   |                     |     |
| Operator's name:                                                                                                               |   |                                         |     | Oiler's name:                 |   |                     |     |
| INSTRUCTIONS: Check all items indicated. Inspect and indicate as satisfactory = S, Unsatisfactory = U, or not applicable = N/A |   |                                         |     |                               |   |                     |     |
| Walk around inspection                                                                                                         | U | S                                       | N/A | Operator Cab Inspection       | U | S                   | N/A |
| Safety guards and plates                                                                                                       |   |                                         |     | Gauges                        |   |                     |     |
| Carrier frame, rotate base                                                                                                     |   |                                         |     | Warning & indicator lights    |   |                     |     |
| General hardware                                                                                                               |   |                                         |     | Control/brakes                |   |                     |     |
| Wire rope                                                                                                                      |   |                                         |     | Visibility                    |   |                     |     |
| Reeving                                                                                                                        |   |                                         |     | Load rating charts            |   |                     |     |
| Block                                                                                                                          |   |                                         |     | Safety devices                |   |                     |     |
| Hook                                                                                                                           |   |                                         |     | Emergency stops               |   |                     |     |
| Sheeves                                                                                                                        |   |                                         |     | List/trim indicators          |   |                     |     |
| Boom/Jib                                                                                                                       |   |                                         |     | Boom Angle/Radius Indicator   |   |                     |     |
| Gantry, pendants, boom stops                                                                                                   |   |                                         |     | Machinery House Inspection    | U | S                   | N/A |
| Walks, ladders, handrails                                                                                                      |   |                                         |     | Housekeeping                  |   |                     |     |
| Wind locks, chocks, stops                                                                                                      |   |                                         |     | Engine/Compressor             |   |                     |     |
| Tires, wheels, tracks                                                                                                          |   |                                         |     | Leaks -Fuel, lube, Oil, Water |   |                     |     |
| Leaks-Fuel, oil, lube, water                                                                                                   |   |                                         |     | Lubrication                   |   |                     |     |
| Radius indicator                                                                                                               |   |                                         |     | Battery                       |   |                     |     |
| Outrigger/locking device                                                                                                       |   |                                         |     | Lights                        |   |                     |     |
| Operation Inspection                                                                                                           | U | S                                       | N/A | Glass                         |   |                     |     |
| Area safety                                                                                                                    |   |                                         |     | Clutch/Brake linings          |   |                     |     |
| Unusual noises                                                                                                                 |   |                                         |     | Electric motors               |   |                     |     |
| Control Action                                                                                                                 |   |                                         |     | Warning tags                  |   |                     |     |
| Brakes/boom/load/ rotate                                                                                                       |   |                                         |     | Fire extinguisher             |   |                     |     |
| Crane stability                                                                                                                |   |                                         |     | Comments:                     |   |                     |     |
| No load test                                                                                                                   |   |                                         |     |                               |   |                     |     |
| Fleeting sheeve                                                                                                                |   |                                         |     |                               |   |                     |     |
| Limit switches                                                                                                                 |   |                                         |     |                               |   |                     |     |
| Operator's signature:                                                                                                          |   |                                         |     | Supervisor's signature:       |   |                     |     |

FSAF-06

## EXCAVATION PERMIT

Number.....

Excavation permits can be used where there are likely to be underground services or where the type / depth of the excavation will involve high risks that need to be controlled.

The type of operation for which this Permit could be used includes:

Excavation Work around live services  
Deep excavations where men have to work  
Contaminated ground  
Excavation in made up ground or in poor soil

### PART A – ISSUE

Permit is issued to: .....

of:.....

for (specific work to be done).....

.....

at (exact location of work) .....

.....

Valid from:..... am/pm

to:..... am/pm Date:.....20.....

**I issue this permit subject to the minimum precautions I have ticked in Part D of this permit above.  
I have listed in Part B below.**

**Signed:** .....  
(Issuer)

### PART D - COMPLETION OF WORK

The work has been completed and:  
the work and adjacent areas have been  
thoroughly inspected to ensure no hazards  
remain

☐

Services re-instated

☐

Route of excavation recorded

☐

Records of disposal of contaminated material  
handed over

☐

All equipment and material removed

☐

Area restored

☐

Other .....

.....

.....

☐

**The work has been completed, and I have  
undertaken the checks and safety measures**

Time:.....am/pm

Date:.....20.....

**PART B - PRECAUTIONS**

- Existing services identified ☐
- Electrical supply isolated ☐
- Temporary trench supports available ☐
- Inspection regime in place ☐
- Traffic control in place ☐
- Protective clothing provided ☐
- Suitable safe access provided ☐
- Location protected and sign marked ☐
- Other ☐

**PART C – ACCEPTANCE**

**I accept this permit and will ensure the precautions ticked above in part B have been undertaken before starting work. I also understand the checks and safety measures I am required to undertake after completion of work.**

Signed: .....(Subcontractor undertaking work)

Signed: .....

☐ (subcontractor once completed work)

**PART E - CANCELLATION OF PERMIT**

**This permit to work is cancelled and the relevant staff have been notified. No more work under this permit.**

Time:.....am/pm

Date: .....20.....

Signed: .....

(Issuer)





## HOT WORK PERMIT

Number.....

Hot work permits are to be used where a significant risk of fire or erroneous triggering of a fire detection system has been identified.

The type of heat, flame or spark producing operation for which this Permit may be used includes:

Welding – Cutting - Grinding - Soldering – Use of Bitumen boilers - Work using LPG – Burning off paint

### PART A – ISSUE

Permit is issued to: .....

for (specific work to be done).....

.....

at (exact location of work) .....

.....

Valid from:..... am/pm

to:..... am/pm Date: .....20.....

**I issue this permit subject to the minimum precautions  
I have listed in Part B below.**

**Signed:** .....

(Issuer)

### PART B - PRECAUTIONS

#### **Before starting:**

Combustible materials: removed ☐  
protected ☐

Electrical equipment/supply isolated ☐

Smoke detector/s: isolated ☐

Extinguisher/s: readily available ☐

Personnel: instructed in fire precautions ☐  
know nearest fire alarm ☐

Location: protected and sign marked ☐

Work area screened to contain sparks ☐

Other ☐

#### **Following completion:**

Work and adjacent areas to be checked immediately ☐

30 minutes later ☐

1 hour after completion ☐

Smoke detector/s to be re-activated ☐

Other ☐

### PART C - ACCEPTANCE

**I accept this permit and will ensure the precautions  
ticked above in part B have been undertaken before  
starting work. I also understand the checks and  
safety measures I am required to undertake after  
Completion of work.**

**Signed:** .....(Subcontractor undertaking work)

### PART D - COMPLETION OF WORK

The work has been completed and the work area and  
adjacent areas to which sparks and heat might have  
spread such as opposite sides of floors and walls)  
have been thoroughly inspected and found fire

immediately following completion ☐

30 minutes later ☐

1 hour after completion ☐

Electrical equipment/supply switched back on ☐

Smoke detector/s have been re-activated ☐

All equipment and material has been removed ☐

Area restored ☐

Other ☐

**The work has been completed, and I have  
undertaken the checks and safety measures  
I have ticked in Part D of this permit above.**

Time:.....am/pm

Date: .....20.....

**Signed:** .....

(Subcontractor once completed work)

☐

☐

☐

☐

☐

☐

☐

☐

☐

☐

☐

### PART E - CANCELLATION OF PERMIT ☐

**This permit to work is cancelled and the relevant  
staff have been notified. No more work under this  
permit.**

Time:.....am/pm

Date: .....20.....

**Signed:** .....

(Issuer)



FSAF 009

## NEAR MISS REPORT

Near miss incidents should be regarded with the same seriousness as personal accidents as they are basically the same. The information recorded below is not to proportion blame (unless a clear disregard of safe working caused the incident) but to enable the company to learn and prevent similar occurrences.

Site/Location:

Contract No.

|                                                       |              |
|-------------------------------------------------------|--------------|
| <b>Near Miss – Please describe what happened:</b>     |              |
| <br><br><br><br>                                      |              |
| <b>Was there an obvious cause of the incident:</b>    |              |
| <br><br><br><br>                                      |              |
| <b>Contributing Factors:</b>                          |              |
| <br><br><br><br>                                      |              |
| <b>Time:</b>                                          | <b>Date:</b> |
| <b>Weather Conditions:</b>                            |              |
| <b>Company Involved:</b>                              |              |
| <b>Personnel Involved:</b>                            |              |
| <b>Plant Equipment Involved:</b>                      |              |
| <b>What has been done to prevent a re-occurrence:</b> |              |
| <br><br><br><br>                                      |              |

Name of Person Making Report.....

Job Title.....

Signature:.....Date.....

Distribution: Area Director, Company Safety Advisor, Sub-Contractor's Office, Notice Boards

Attached: Photos Y / N / NA

Sketches Y / N / NA

Risk Assessments Y / N / NA

Statements Y / N / NA